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CoCa: Contrastive Captioners are Image-Text Foundation Models

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Abstract

Exploring large-scale pretrained foundation models is of significant interest in computer vision because these models can be quickly transferred to many downstream tasks. This paper presents **Co**ntrastive **Ca**ptioner (**CoCa**), a minimalist design to pretrain an image-text encoder-decoder foundation model jointly with contrastive loss and captioning loss, thereby subsuming model capabilities from contrastive approaches like CLIP and generative methods like SimVLM. In contrast to standard encoder-decoder transformers where all decoder layers attend to encoder outputs, CoCa omits cross-attention in the first half of decoder layers to encode *unimodal* text representations, and cascades the remaining decoder layers which cross-attend to the image encoder for *multimodal* image-text representations. We apply a contrastive loss between unimodal image and text embeddings, in addition to a captioning loss on the multimodal decoder outputs which predicts text tokens autoregressively. By sharing the same computational graph, the two training objectives are computed efficiently with minimal overhead. CoCa is pretrained end-to-end and from scratch on both web-scale alt-text data and annotated images by treating all labels simply as text, seamlessly unifying natural language supervision for representation learning. Empirically, CoCa achieves state-of-the-art performance with zero-shot transfer or minimal task-specific adaptation on a broad range of downstream tasks, spanning visual recognition (ImageNet, Kinetics-400/600/700, Moments-in-Time), crossmodal retrieval (MSCOCO, Flickr30K, MSR-VTT), multimodal understanding (VQA, SNLI-VE, NLVR2), and image captioning (MSCOCO, NoCaps). Notably on ImageNet classification, CoCa obtains 86.3% *zero-shot* top-1 accuracy, 90.6% with a *frozen encoder* and learned classification head, and new state-of-the-art 91.0% top-1 accuracy on ImageNet with a *finetuned encoder*.

1 Introduction

Deep learning has recently witnessed the rise of foundation language models [1] such as BERT [2], T5 [3], GPT-3 [4], where models are pretrained on web-scale data and demonstrate generic multi-tasking capabilities through zero-shot, few-shot or transfer learning. Compared with specialized individual models, pretraining foundation models for massive downstream tasks can amortize training costs, providing opportunities to push the limits of model scale [5] for human-level intelligence.

For vision and vision-language problems, several foundation model candidates have been explored: (1) Pioneering works [6, 7, 8] have shown the effectiveness of **single-encoder models** pretrained with cross-entropy loss on image classification datasets such as ImageNet [9]. The image encoder provides generic *visual representations* that can be adapted for various downstream tasks including

[†]Equal contribution.

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CoCa：对比式描述生成器是图像-文本基础模型

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摘要

探索大规模预训练基础模型在计算机视觉领域具有重要意义，因为这些模型可以快速迁移到许多下游任务。本文提出了 Contrastive Captioner（CoCa），一种极简设计，用于通过对比损失和图像描述生成损失联合预训练图像-文本编码器-解码器基础模型，从而兼具 CLIP 等对比方法和 SimVLM 等生成方法的模型能力。与所有解码器层都会关注编码器输出的标准编码器-解码器 Transformer 不同，CoCa 在前半部分解码器层中省略了交叉注意力，以编码单模态文本表示，并级联剩余的解码器层，使其对图像编码器进行交叉注意力，以获得多模态图像-文本表示。除了在多模态解码器输出上施加用于自回归预测文本标记的图像描述生成损失外，我们还在单模态图像和文本嵌入之间应用对比损失。通过共享同一计算图，这两个训练目标能够以极小的额外开销高效计算。CoCa 在网络规模的替代文本数据和带标注图像上以端到端方式从零开始预训练，并将所有标签都简单地视为文本，从而无缝统一了用于表征学习的自然语言监督。实验结果表明，CoCa 在广泛的下游任务上，通过零样本迁移或最少的任务特定适配，取得了最先进的性能，涵盖视觉识别（ImageNet、Kinetics-400/600/700、Moments-in-Time）、跨模态检索（MSCOCO、Flickr30K、MSR-VTT）、多模态理解（VQA、SNLI-VE、NLVR2）和图像描述生成（MSCOCO、NoCaps）。尤其是在 ImageNet 分类任务上，CoCa 获得了 86.3% 的零样本 top-1 准确率、在冻结编码器并学习分类头时获得 90.6% 的准确率，并在微调编码器后在 ImageNet 上取得了新的最先进 91.0% top-1 准确率。

1 引言

深度学习近来见证了基础语言模型 [1] 的兴起，如 BERT [2]、T5 [3]、GPT-3 [4]，这些模型在网络规模的数据上进行预训练，并通过零样本、少样本或迁移学习展现出通用的多任务能力。与专门的单一模型相比，为海量下游任务预训练基础模型可以摊销训练成本，为推动模型规模 [5] 的极限以实现人类水平的智能提供了机会。

对于视觉和视觉-语言问题，已有多种基础模型候选方案被探索：（1）开创性工作 [6, 7, 8] 已经证明了在 ImageNet [9] 等图像分类数据集上使用交叉熵损失预训练的单编码器模型的有效性。图像编码器提供了通用的视觉表示，可适用于包括以下内容在内的各种下游任务

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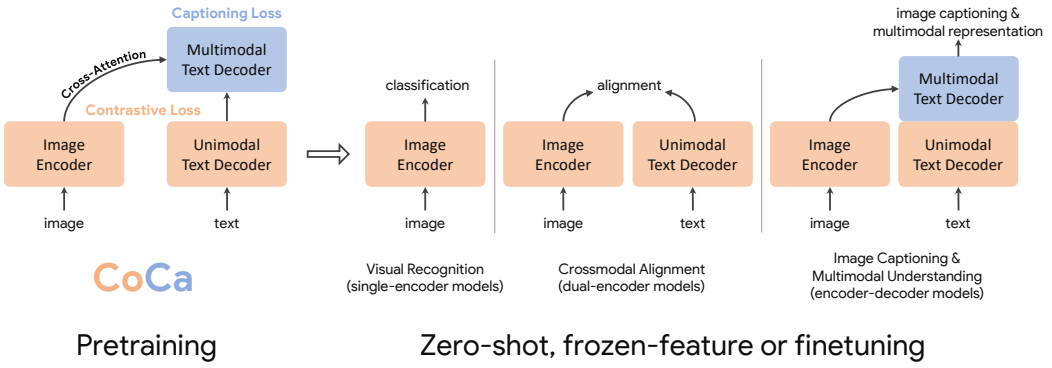


Figure 1: Overview of Contrastive Captioners (CoCa) pretraining as image-text foundation models. The pretrained CoCa can be used for downstream tasks including visual recognition, vision-language alignment, image captioning and multimodal understanding with zero-shot transfer, frozen-feature evaluation or end-to-end finetuning.

image and video understanding [10, 11]. However, these models rely heavily on image annotations as labeled vectors and do not bake in knowledge of free-form human natural language, hindering their application to downstream tasks that involving both vision and language modalities. (2) Recently, a line of research [12, 13, 14] has shown the feasibility of image-text foundation model candidates by pretraining two parallel encoders with a contrastive loss on web-scale noisy image-text pairs. In addition to the visual embeddings for vision-only tasks, the resulting **dual-encoder models** can additionally encode textual embeddings to the same latent space, enabling new *crossmodal alignment* capabilities such as zero-shot image classification and image-text retrieval. Nonetheless, these models are not directly applicable for joint vision-language understanding tasks such as visual question answering (VQA), due to missing joint components to learn fused image and text representations. (3) Another line of research [15, 16, 17] has explored generative pretraining with **encoder-decoder models** to learn generic vision and multimodal representations. During pretraining, the model takes images on the encoder side and applies Language Modeling (LM) loss (or PrefixLM [3, 16]) on the decoder outputs. For downstream tasks, the decoder outputs can then be used as joint representations for *multimodal understanding* tasks. While superior vision-language results [16] have been attained with pretrained encoder-decoder models, they do not produce text-only representations aligned with image embeddings, thereby being less feasible and efficient for crossmodal alignment tasks.

In this work, we unify *single-encoder*, *dual-encoder* and *encoder-decoder* paradigms, and train one image-text foundation model that subsumes the capabilities of all three approaches. We propose a simple model family named Contrastive Captioners (CoCa) with a modified encoder-decoder architecture trained with both contrastive loss and captioning (generative) loss. As shown in Figure 1, we decouple the decoder transformer into two parts, a unimodal decoder and a multimodal decoder. We omit cross-attention in unimodal decoder layers to encode text-only representations, and cascade multimodal decoder layers cross-attending to image encoder outputs to learn *multimodal* image-text representations. We apply both the contrastive objective between outputs of the image encoder and unimodal text decoder, and the captioning objective at the output of the multimodal decoder. Furthermore, CoCa is trained on both image annotation data and noisy image-text data by treating all labels simply as text. The generative loss on image annotation text provides a fine-grained training signal similar to the single-encoder cross-entropy loss approach, effectively subsuming all three pretraining paradigms into a single unified method.

The design of CoCa leverages contrastive learning for learning global representations and captioning for fine-grained region-level features, thereby benefiting tasks across all three categories shown in Figure 1. CoCa shows that a single pretrained model can outperform many specialized models using zero-shot transfer or minimal task-specific adaptation. For example, CoCa obtains 86.3% zero-shot accuracy on ImageNet and better zero-shot crossmodal retrieval on MSCOCO and Flickr30k. With a frozen-encoder, CoCa achieves 90.6% on ImageNet classification, 88.0%/88.5%/81.1% on Kinetics-40/600/700 and 47.4% on Moments-in-Time. After lightweight finetuning, CoCa further achieves 91.0% on ImageNet, 82.3% on VQA and 120.6 CIDEr score on NoCaps.

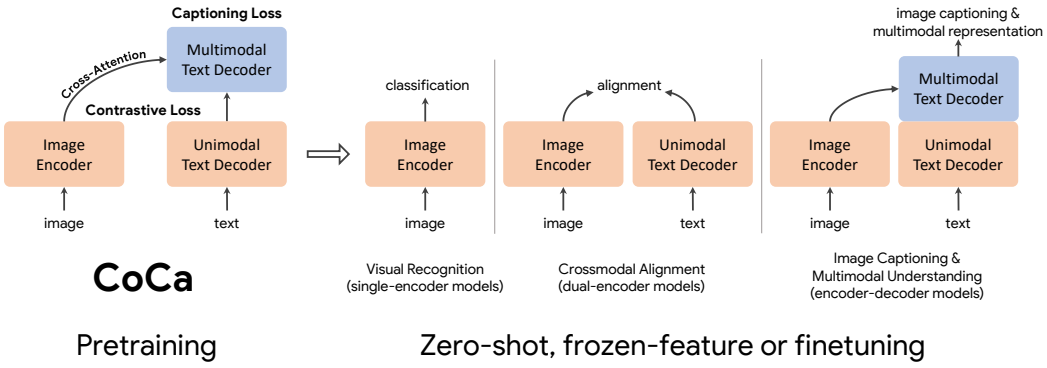


图 1：作为图文基础模型的对比式图像描述器（CoCa）预训练概览。预训练的 CoCa 可用于下游任务，包括视觉识别、视觉-语言对齐、图像描述生成和多模态理解，并支持零样本迁移、冻结特征评估或端到端微调。

图像和视频理解 [10, 11]。然而，这些模型严重依赖作为标注向量的图像注释，并未内化自由形式的人类自然语言知识，从而阻碍了它们在同时涉及视觉和语言模态的下游任务中的应用。（2）近期，一系列研究 [12, 13, 14] 通过在网络规模的噪声图像-文本对上使用对比损失预训练两个并行编码器，展示了图像-文本基础模型候选方案的可行性。除了用于纯视觉任务的视觉嵌入外，所得的双编码器模型还可以将文本嵌入编码到同一潜在空间中，从而实现新的跨模态对齐能力，如零样本图像分类和图像-文本检索。尽管如此，由于缺少用于学习融合图像和文本表示的联合组件，这些模型无法直接应用于视觉问答（VQA）等联合视觉-语言理解任务。（3）另一系列研究 [15, 16, 17] 探索了使用编码器-解码器模型进行生成式预训练，以学习通用视觉和多模态表示。在预训练过程中，模型在编码器端接收图像，并在解码器输出上应用语言建模（LM）损失（或 PrefixLM [3, 16]）。对于下游任务，解码器输出随后可用作多模态理解任务的联合表示。尽管预训练的编码器-解码器模型已经取得了优越的视觉-语言结果 [16]，但它们不会生成与图像嵌入对齐的纯文本表示，因此对于跨模态对齐任务而言可行性和效率较低。

在这项工作中，我们统一了单编码器、双编码器和编码器-解码器范式，并训练了一个图像-文本基础模型，该模型涵盖了这三种方法的能力。我们提出了一个名为对比式字幕生成器（Contrastive Captioners，CoCa）的简单模型家族，其采用经过修改的编码器-解码器架构，并同时使用对比损失和字幕生成（生成式）损失进行训练。如图 1 所示，我们将解码器 Transformer 解耦为两部分：单模态解码器和多模态解码器。我们在单模态解码器层中省略交叉注意力，以编码纯文本表征，并级联多模态解码器层，使其对图像编码器输出进行交叉注意力，从而学习多模态图像-文本表征。我们同时应用图像编码器输出与单模态文本解码器输出之间的对比目标，以及多模态解码器输出处的字幕生成目标。此外，CoCa 通过将所有标签简单地视为文本，在图像标注数据和有噪声的图像-文本数据上进行训练。图像标注文本上的生成式损失提供了类似于单编码器交叉熵损失方法的细粒度训练信号，从而有效地将所有三种预训练范式纳入一个统一的方法中。

CoCa 的设计利用对比学习来学习全局表征，并利用图像描述生成来学习细粒度的区域级特征，从而使图 1 所示的三类任务都受益。CoCa 表明，单个预训练模型可以通过零样本迁移或最少的任务特定适配，超越许多专用模型。例如，CoCa 在 ImageNet 上取得了 86.3% 的零样本准确率，并在 MSCOCO 和 Flickr30k 上实现了更好的零样本跨模态检索。使用冻结编码器时，CoCa 在 ImageNet 分类上达到 90.6%，在 Kinetics-40/600/700 上分别达到 88.0%/88.5%/81.1%，在 Moments-in-Time 上达到 47.4%。经过轻量级微调后，CoCa 进一步在 ImageNet 上达到 91.0%，在 VQA 上达到 82.3%，并在 NoCaps 上取得 120.6 的 CIDEr 分数。

2 Related Work

Vision Pretraining. Pretraining ConvNets [18] or Transformers [19] on large-scale annotated data such as ImageNet [6, 7, 8], Instagram [20] or JFT [21] has become a popular strategy towards solving visual recognition problems including classification, localization, segmentation, video recognition, tracking and many other problems. Recently, self-supervised pretraining approaches have also been explored. BEiT [22] proposes a masked image modeling task following BERT [2] in natural language processing, and uses quantized visual token ids as prediction targets. MAE [23] and SimMIM [24] remove the need for an image tokenizer and directly use a light-weight decoder or projection layer to regress pixel values. Nonetheless, these methods only learn models for the vision modality and thus they are not applicable to tasks that require joint reasoning over both image and text inputs.

Vision-Language Pretraining. In recent years, rapid progress has been made in vision-language pretraining (VLP), which aims to jointly encode vision and language in a fusion model. Early work (e.g. LXMERT [25], UNITER [26], VinVL [27]) in this direction relies on pretrained object detection modules such as Fast(er) R-CNN [28] to extract visual representations. Later efforts such as ViLT [29] and VLMo [30] unify vision and language transformers, and train a multimodal transformer from scratch.

Image-Text Foundation Models. Recent work has proposed image-text foundation models that can subsume both vision and vision-language pretraining. CLIP [12] and ALIGN [13] demonstrate that dual-encoder models pretrained with contrastive objectives on noisy image-text pairs can learn strong image and text representations for crossmodal alignment tasks and zero-shot image classification. Florence [14] further develops this method with unified contrastive objective [31], training foundation models that can be adapted for a wide range of vision and image-text benchmarks. To further improve zero-shot image classification accuracy, LiT [32] and BASIC [33] first pretrain model on an large-scale image annotation dataset with cross-entropy and further finetune with contrastive loss on an noisy alt-text image dataset. Another line of research [16, 17, 34] proposes encoder-decoder models trained with generative losses and shows strong results in vision-language benchmarks while the visual encoder still performs competitively on image classification. In this work, we focus on training an image-text foundation model from scratch in a single pretraining stage to unify these approaches. While recent works [35, 36, 37] have also explored image-text unification, they require multiple pretraining stages of unimodal and multimodal modules to attain good performance. For example, ALBEF [36] combines contrastive loss with masked language modelling (MLM) with a dual-encoder design. However, our approach is simpler and more efficient to train while also enables more model capabilities: (1) CoCa only performs one forward and backward propagation for a batch of image-text pairs while ALBEF requires two (one on corrupted inputs and another without corruption), (2) CoCa is trained from scratch on the two objectives only while ALBEF is initialized from pretrained visual and textual encoders with additional training signals including momentum modules. (3) The decoder architecture with generative loss is preferred for natural language generation and thus directly enables image captioning and zero-shot learning [16].

3 Approach

We begin with a review of three foundation model families that utilize *natural language supervision* differently: single-encoder classification pretraining, dual-encoder contrastive learning, and encoder-decoder image captioning. We then introduce Contrastive Captioners (CoCa) that share the merits of both contrastive learning and image-to-caption generation under a simple architecture. We further discuss how CoCa models can quickly transfer to downstream tasks with zero-shot transfer or minimal task adaptation.

3.1 Natural Language Supervision

Single-Encoder Classification. The classic single-encoder approach pretrains a visual encoder through image classification on a large crowd-sourced image annotation dataset (*e.g.*, ImageNet [9], Instagram [20] or JFT [21]), where the vocabulary of annotation texts is usually fixed. These image annotations are usually mapped into discrete class vectors to learn with a cross-entropy loss as

$$\mathcal{L}_{\text{Cls}} = -p(y) \log q_\theta(x), \tag{1}$$

2 相关工作

视觉预训练。在大规模标注数据（如 ImageNet [6, 7, 8]、Instagram [20] 或 JFT [21]）上预训练 ConvNets [18] 或 Transformers [19]，已成为解决包括分类、定位、分割、视频识别、跟踪以及许多其他问题在内的视觉识别问题的一种流行策略。近来，自监督预训练方法也得到了探索。BEiT [22] 借鉴自然语言处理中的 BERT [2]，提出了一种掩码图像建模任务，并使用量化的视觉 token id 作为预测目标。MAE [23] 和 SimMIM [24] 则不再需要图像分词器，而是直接使用轻量级解码器或投影层来回归像素值。尽管如此，这些方法仅学习视觉模态的模型，因此不适用于需要对图像和文本输入进行联合推理的任务。

视觉-语言预训练。近年来，视觉-语言预训练（VLP）取得了快速进展，其目标是在融合模型中联合编码视觉和语言。该方向的早期工作（例如 LXMERT [25]、UNITER [26]、VinVL [27]）依赖于预训练的目标检测模块，如 Fast(er) R-CNN [28]，来提取视觉表示。后续工作如 ViLT [29] 和 VLMo [30] 统一了视觉与语言 Transformer，并从头开始训练多模态 Transformer。

图文基础模型。近期研究提出了图文基础模型，能够同时涵盖视觉预训练和视觉-语言预训练。CLIP [12] 和 ALIGN [13] 表明，在有噪声的图文对上使用对比目标进行预训练的双编码器模型，可以为跨模态对齐任务和零样本图像分类学习到强大的图像与文本表征。Florence [14] 通过统一的对比目标 [31] 进一步发展了这一方法，训练出可适配广泛视觉和图文基准的基础模型。为进一步提升零样本图像分类准确率，LiT [32] 和 BASIC [33] 首先在大规模图像标注数据集上使用交叉熵预训练模型，然后在有噪声的替代文本图像数据集上使用对比损失进一步微调。另一条研究路线 [16, 17, 34] 提出了使用生成式损失训练的编码器-解码器模型，并在视觉-语言基准上取得了强劲结果，同时其视觉编码器在图像分类上仍具有竞争力。在本文中，我们专注于在单一预训练阶段从零开始训练图文基础模型，以统一这些方法。尽管近期工作 [35, 36, 37] 也探索了图文统一，但它们需要对单模态和多模态模块进行多个预训练阶段，才能获得良好性能。例如，ALBEF [36] 采用双编码器设计，将对比损失与掩码语言建模（MLM）相结合。然而，我们的方法更简单且训练效率更高，同时还支持更多模型能力：（1）对于一批图文对，CoCa 只执行一次前向和反向传播，而 ALBEF 需要两次（一次使用被破坏的输入，另一次使用未破坏的输入）；（2）CoCa 仅基于两个目标从零开始训练，而 ALBEF 则从预训练的视觉和文本编码器初始化，并使用包括动量模块在内的额外训练信号；（3）带有生成式损失的解码器架构更适合自然语言生成，因此可直接支持图像描述生成和零样本学习 [16]。

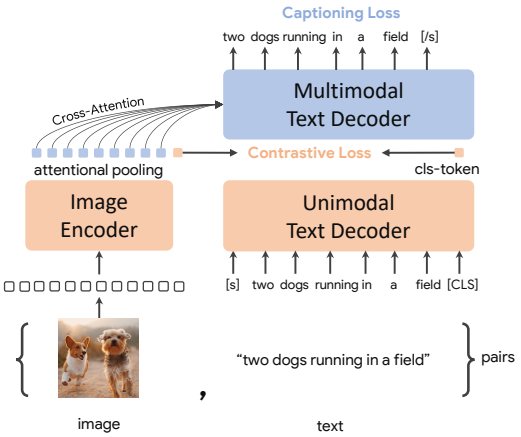
3 方法

我们首先回顾三类以不同方式利用自然语言监督的基础模型家族：单编码器分类预训练、双编码器对比学习，以及编码器-解码器图像描述生成。随后，我们介绍对比式描述器（CoCa），它在一种简单架构下兼具对比学习和图像到描述生成的优点。我们进一步讨论 CoCa 模型如何通过零样本迁移或最小化任务适配，快速迁移到下游任务。

3.1 自然语言监督

单编码器分类。经典的单编码器方法通过在大规模众包图像标注数据集（例如 ImageNet [9]、Instagram [20] 或 JFT [21]）上进行图像分类来预训练视觉编码器，其中标注文本的词汇表通常是固定的。这些图像标注通常被映射为离散的类别向量，并使用交叉熵损失进行学习，其形式为：

$$\mathcal{L}_{\text{Cls}} = -p(y) \log q_\theta(x), \tag{1}$$



Algorithm 1 Pseudocode of Contrastive Captioners architecture.

```
# image, text.ids, text.labels, text.mask: paired {image, text} data
# con_query: 1 query token for contrastive embedding
# cap_query: N query tokens for captioning embedding
# cls_token_id: a special cls_token_id in vocabulary

def attentional_pooling(features, query):
  out = multihead_attention(features, query)
  return layer_norm(out)

img_feature = vit_encoder(image) # [batch, seq_len, dim]
con_feature = attentional_pooling(img_feature, con_query) # [batch, 1, dim]
cap_feature = attentional_pooling(img_feature, cap_query) # [batch, N, dim]

ids = concat(text.ids, cls_token_id)
mask = concat(text.mask, zeros_like(cls_token_id)) # unpad cls_token_id
txt_embs = embedding_lookup(ids)
unimodal_out = lm_transformers(txt_embs, mask, cross_attn=None)
multimodal_out = lm_transformers(
    unimodal_out[:, :-1, :], mask, cross_attn=cap_feature)
cls_token_feature = layer_norm(unimodal_out)[:, -1:, :] # [batch, 1, dim]

con_loss = contrastive_loss(con_feature, cls_token_feature)
cap_loss = softmax_cross_entropy_loss(
    multimodal_out, labels=text.labels, mask=text.mask)
```

`vit_encoder`: vision transformer based encoder; `lm_transformer`: language-model transformers.

Figure 2: Detailed illustration of CoCa architecture and training objectives.

where $p(y)$ is a one-hot, multi-hot or smoothed label distribution from ground truth label y. The learned image encoder is then used as a generic visual representation extractor for downstream tasks.

Dual-Encoder Contrastive Learning. Compared to pretraining with single-encoder classification, which requires human-annotated labels and data cleaning, the dual-encoder approach exploits noisy web-scale text descriptions and introduces a learnable text tower to encode free-form texts. The two encoders are jointly optimized by contrasting the paired text against others in the sampled batch:

$$\mathcal{L}_{\text{Con}} = -\frac{1}{N}\Big(\underbrace{\sum_{i}^{N} \log \frac{\exp(x_i^\top y_i/\sigma)}{\sum_{j=1}^{N}\exp(x_i^\top y_j/\sigma)}}_{\text{image-to-text}} + \underbrace{\sum_{i}^{N} \log \frac{\exp(y_i^\top x_i/\sigma)}{\sum_{j=1}^{N}\exp(y_i^\top x_j/\sigma)}}_{\text{text-to-image}}\Big), \tag{2}$$

where x_i and y_j are normalized embeddings of the image in the i-th pair and that of the text in the j-th pair. N is the batch size, and σ is the temperature to scale the logits. In addition to the image encoder, the dual-encoder approach also learns an aligned text encoder that enables crossmodal alignment applications such as image-text retrieval and zero-shot image classification. Empirical evidence shows zero-shot classification is more robust [12, 13, 38] on corrupted or out-of-distribution images.

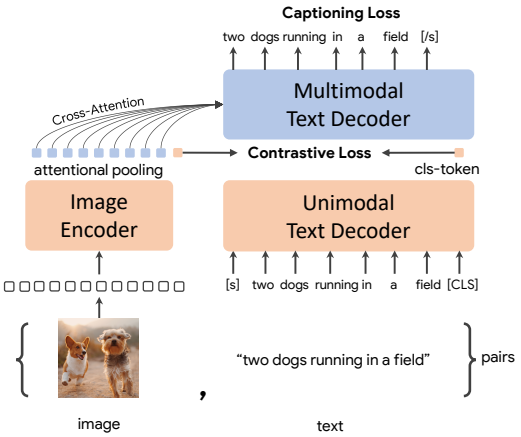
Encoder-Decoder Captioning. While the dual-encoder approach encodes the text as a whole, the generative approach (*a.k.a.* captioner) aims for detailed granularity and requires the model to predict the exact tokenized texts of y autoregressively. Following a standard encoder-decoder architecture, the image encoder provides latent encoded features (*e.g.*, using a Vision Transformer [39] or ConvNets [40]) and the text decoder learns to maximize the conditional likelihood of the paired text y under the forward autoregressive factorization:

$$\mathcal{L}_{\text{Cap}} = -\sum_{t=1}^{T} \log P_\theta(y_t | y_{<t}, x). \tag{3}$$

The encoder-decoder is trained with teacher-forcing [41] to parallelize computation and maximize learning efficiency. Unlike prior methods, the captioner approach yields a joint image-text representation that can be used for vision-language understanding, and is also capable of image captioning applications with natural language generation.

3.2 Contrastive Captioners Pretraining

Figure 2 depicts the proposed contrastive captioner (CoCa): a simple encoder-decoder approach that seamlessly combines the three training paradigms. Similar to standard image-text encoder-decoder models, CoCa encodes images to latent representations by a neural network encoder, for example, vision transformer (ViT) [39] (used by default; it can also be other image encoders like ConvNets [40]), and decodes texts with a causal masking transformer decoder. Unlike standard



Algorithm 1 Pseudocode of Contrastive Captioners architecture.

```
# image, text.ids, text.labels, text.mask: paired {image, text} data
# con_query: 1 query token for contrastive embedding
# cap_query: N query tokens for captioning embedding
# cls_token_id: a special cls_token_id in vocabulary

def attentional_pooling(features, query):
  out = multihead_attention(features, query)
  return layer_norm(out)

img_feature = vit_encoder(image) # [batch, seq_len, dim]
con_feature = attentional_pooling(img_feature, con_query) # [batch, 1, dim]
cap_feature = attentional_pooling(img_feature, cap_query) # [batch, N, dim]

ids = concat(text.ids, cls_token_id)
mask = concat(text.mask, zeros_like(cls_token_id)) # unpad cls_token_id
txt_embs = embedding_lookup(ids)
unimodal_out = lm_transformers(txt_embs, mask, cross_attn=None)
multimodal_out = lm_transformers(
    unimodal_out[:, :-1, :], mask, cross_attn=cap_feature)
cls_token_feature = layer_norm(unimodal_out)[:, -1:, :] # [batch, 1, dim]

con_loss = contrastive_loss(con_feature, cls_token_feature)
cap_loss = softmax_cross_entropy_loss(
    multimodal_out, labels=text.labels, mask=text.mask)
```

vit_encoder: vision transformer based encoder; lm_transformer: language-model transformers.

图 2：CoCa 架构和训练目标的详细示意图。

其中，$p(y)$ 是来自真实标签 y 的 one-hot、multi-hot 或平滑标签分布。随后，学习得到的图像编码器被用作下游任务的通用视觉表示提取器。

双编码器对比学习。与需要人工标注标签和数据清洗的单编码器分类预训练相比，双编码器方法利用噪声较大的网络规模文本描述，并引入一个可学习的文本塔来编码自由形式文本。两个编码器通过将配对文本与采样批次中的其他文本进行对比来联合优化：

$$\mathcal{L}_{\text{Con}} = -\frac{1}{N}\Big(\underbrace{\sum_{i}^{N}\log\frac{\exp(x_i^\top y_i/\sigma)}{\sum_{j=1}^{N}\exp(x_i^\top y_j/\sigma)}}_{\text{image-to-text}} + \underbrace{\sum_{i}^{N}\log\frac{\exp(y_i^\top x_i/\sigma)}{\sum_{j=1}^{N}\exp(y_i^\top x_j/\sigma)}}_{\text{text-to-image}}\Big), \tag{2}$$

其中，x_i 和 y_j 分别是第 i 个配对中的图像和第 j 个配对中的文本的归一化嵌入。N 是批量大小，σ 是用于缩放 logits 的温度。除了图像编码器之外，双编码器方法还学习一个对齐的文本编码器，从而支持图像-文本检索和零样本图像分类等跨模态对齐应用。经验证据表明，在损坏或分布外图像上，零样本分类更加鲁棒 [12, 13, 38]。

编码器-解码器式字幕生成。虽然双编码器方法将文本作为整体进行编码，但生成式方法（又称字幕生成器）旨在实现更细粒度的建模，并要求模型以自回归方式预测 y 的精确分词文本。遵循标准的编码器-解码器架构，图像编码器提供潜在的编码特征（例如，使用视觉 Transformer [39] 或卷积网络 [40]），文本解码器则学习在前向自回归分解下最大化配对文本 y 的条件似然：

$$\mathcal{L}_{\text{Cap}} = -\sum_{t=1}^{T}\log P_\theta(y_t|y_{<t}, x). \tag{3}$$

编码器-解码器采用教师强制 [41] 进行训练，以并行化计算并最大化学习效率。与以往方法不同，图像描述生成器方法能够产生可用于视觉-语言理解的联合图像-文本表征，同时也具备通过自然语言生成进行图像描述的应用能力。

3.2 对比式图注生成器预训练

图 2 展示了所提出的对比式字幕生成器（CoCa）：一种简单的编码器-解码器方法，能够无缝结合三种训练范式。与标准的图像-文本编码器-解码器模型类似，CoCa 通过神经网络编码器将图像编码为潜在表示，例如视觉 Transformer（ViT）[39]（默认使用；也可以是其他图像编码器，如 ConvNets [40]），并使用因果掩码 Transformer 解码器解码文本。不同于标准的

	Image Encoder			Text Decoder				Image / Text		
Model	Layers	MLP	Params	n_{uni}	n_{multi}	MLP	Params	Hidden	Heads	Total Params
CoCa-Base	12	3072	86M	12	12	3072	297M	768	12	383M
CoCa-Large	24	4096	303M	12	12	4096	484M	1024	16	787M
CoCa	40	6144	1B	18	18	5632	1.1B	1408	16	2.1B

Table 1: Size variants of CoCa. Both image encoder and text decoder are Transformers [19, 39].

decoder transformers, CoCa omits cross-attention in the first half of the decoder layers to encode *unimodal* text representations, and cascades the rest of the decoder layers, cross-attending to the image encoder for *multimodal* image-text representations. As a result, the CoCa decoder simultaneously produces both unimodal and multimodal text representations that allow us to apply both contrastive and generative objectives as

$$\mathcal{L}_{\text{CoCa}} = \lambda_{\text{Con}} \cdot \mathcal{L}_{\text{Con}} + \lambda_{\text{Cap}} \cdot \mathcal{L}_{\text{Cap}}, \tag{4}$$

where λ_{Con} and λ_{Cap} are loss weighting hyper-parameters. We note that the single-encoder cross-entropy classification objective can be interpreted as a special case of the generative approach applied on image annotation data, when the vocabulary is the set of all label names.

Decoupled Text Decoder and CoCa Architecture. The captioning approach optimizes the conditional likelihood of text while the contrastive approach uses an unconditional text representation. To address this dilemma and combine these two methods into a single model, we propose a simple *decoupled decoder* design where we split the decoder into unimodal and multimodal components, by skipping the cross-attention mechanism in the unimodal decoder layers. That is, the bottom n_{uni} unimodal decoder layers encode the input text as latent vectors with causally-masked self-attention, and the top n_{multi} multimodal layers further apply causally-masked self-attention and together with cross-attention to the output of the visual encoder. All decoder layers prohibit tokens from attending to future tokens, and it is straightforward to use the multimodal text decoder output for the captioning objective $\mathcal{L}_{\text{Cap}}$. For the contrastive objective $\mathcal{L}_{\text{Con}}$, we append a learnable `[CLS]` token at the end of the input sentence and use its corresponding output of unimodal decoder as the text embedding. We split the decoder in half such that $n_{\text{uni}} = n_{\text{multi}}$. Following ALIGN [13], we pretrain with image resolution of 288×288 and patch size 18×18, resulting in a total of 256 image tokens. Our largest CoCa model ("CoCa" in short) follows the ViT-giant setup in [21] with 1B-parameters in the image encoder and 2.1B-parameters altogether with the text decoder. We also explore two smaller variants of "CoCa-Base" and "CoCa-Large" detailed in Table 1.

Attentional Poolers. It is noteworthy that the contrastive loss uses a single embedding for each image while the decoder usually attends to a sequence of image output tokens in an encoder-decoder captioner [16]. Our preliminary experiments show that a single pooled image embedding helps visual recognition tasks as a global representation, while more visual tokens (thus more fine-grained) are beneficial for multimodal understanding tasks which require region-level features. Hence, CoCa adopts task-specific attentional pooling [42] to customize visual representations to be used for different types of training objectives and downstream tasks. Here, a *pooler* is a single multi-head attention layer with n_{query} learnable queries, with the encoder output as both keys and values. Through this, the model can learn to pool embeddings with different lengths for the two training objectives, as shown in Figure 2. The use of task-specific pooling not only addresses different needs for different tasks but also introduces the pooler as a natural task adapter. We use attentional poolers in pretraining for generative loss $n_{\text{query}} = 256$ and contrastive loss $n_{\text{query}} = 1$.

Pretraining Efficiency. A key benefit of the decoupled autoregressive decoder design is that it can compute two training losses considered efficiently. Since unidirectional language models are trained with causal masking on complete sentences, the decoder can efficiently generate outputs for both contrastive and generative losses with a single forward propagation (compared to two passes for a bidirectional approach [36]). Therefore, the majority of the compute is shared between the two losses and CoCa only induces minimal overhead compared to standard encoder-decoder models. On the other hand, while many existing methods [30, 32, 33, 35, 36, 37] train model components with multiple stages on various data sources and/or modalities, CoCa is pretrained end-to-end from scratch directly with various data sources (*i.e.*, annotated images and noisy alt-text images) by treating all labels as texts for both contrastive and generative objectives.

	Image Encoder			Text Decoder				Image / Text		
Model	Layers	MLP	Params	n_{uni}	n_{multi}	MLP	Params	Hidden	Heads	Total Params
CoCa-Base	12	3072	86M	12	12	3072	297M	768	12	383M
CoCa-Large	24	4096	303M	12	12	4096	484M	1024	16	787M
CoCa	40	6144	1B	18	18	5632	1.1B	1408	16	2.1B

表 1：CoCa 的规模变体。图像编码器和文本解码器均为 Transformer [19, 39]。

解码器 Transformer 中，CoCa 在解码器层的前半部分省略交叉注意力，以编码单模态文本表示，并将其余解码器层级联起来，通过对图像编码器进行交叉注意力来获得多模态图像-文本表示。因此，CoCa 解码器同时产生单模态和多模态文本表示，使我们能够同时应用对比式和生成式目标，如下所示：

$$\mathcal{L}_{\text{CoCa}} = \lambda_{\text{Con}} \cdot \mathcal{L}_{\text{Con}} + \lambda_{\text{Cap}} \cdot \mathcal{L}_{\text{Cap}}, \tag{4}$$

其中 λ_{Con} 和 λ_{Cap} 是损失加权超参数。我们注意到，当词汇表为所有标签名称的集合时，单编码器交叉熵分类目标可以解释为生成式方法应用于图像标注数据的一个特例。

解耦文本解码器与 CoCa 架构。图像描述方法优化文本的条件似然，而对比方法使用无条件的文本表示。为了解决这一困境并将这两种方法结合到单一模型中，我们提出了一种简单的解耦解码器设计：通过在单模态解码器层中跳过交叉注意力机制，将解码器拆分为单模态组件和多模态组件。也就是说，底部的 n_{uni} 个单模态解码器层使用因果掩码自注意力将输入文本编码为潜在向量，而顶部的 n_{multi} 个多模态层进一步应用因果掩码自注意力，并结合对视觉编码器输出的交叉注意力。所有解码器层都禁止词元关注未来词元，并且可以直接使用多模态文本解码器输出作为图像描述目标 $\mathcal{L}_{\text{Cap}}$。对于对比目标 $\mathcal{L}_{\text{Con}}$，我们在输入句子的末尾附加一个可学习的 [CLS] 词元，并使用其对应的单模态解码器输出作为文本嵌入。我们将解码器对半拆分，使得 $n_{\text{uni}} = n_{\text{multi}}$。遵循 ALIGN [13]，我们以 288×288 的图像分辨率和 18×18 的补丁大小进行预训练，从而总共产生 256 个图像词元。我们最大的 CoCa 模型（简称“CoCa”）遵循 [21] 中的 ViT-giant 设置，图像编码器包含 10 亿参数，与文本解码器合计共有 21 亿参数。我们还探索了两个较小的变体：“CoCa-Base”和“CoCa-Large”，详见表 1。

注意力池化器。值得注意的是，对比损失对每张图像使用单个嵌入，而在编码器-解码器式图像描述生成器中，解码器通常会关注一系列图像输出标记 [16]。我们的初步实验表明，单个池化后的图像嵌入作为全局表示有助于视觉识别任务，而更多视觉标记（因此更细粒度）有利于需要区域级特征的多模态理解任务。因此，CoCa 采用任务特定的注意力池化 [42]，以定制用于不同类型训练目标和下游任务的视觉表示。在这里，池化器是一个单层多头注意力层，包含 n_{query} 个可学习查询，并将编码器输出同时作为键和值。通过这种方式，模型可以学习为两个训练目标池化不同长度的嵌入，如图 2 所示。使用任务特定的池化不仅满足了不同任务的不同需求，还将池化器作为一种自然的任务适配器引入。我们在预训练中使用注意力池化器：生成损失 $n_{\text{query}} = 256$，对比损失 $n_{\text{query}} = 1$。

预训练效率。解耦自回归解码器设计的一个关键优势在于，它能够高效地计算所考虑的两种训练损失。由于单向语言模型是在完整句子上使用因果掩码进行训练的，解码器可以通过一次前向传播同时为对比损失和生成损失生成输出（相比之下，双向方法需要两次传播 [36]）。因此，两种损失共享了大部分计算量，与标准的编码器-解码器模型相比，CoCa 只引入了极小的额外开销。另一方面，尽管许多现有方法 [30, 32, 33, 35, 36, 37] 在各种数据源和/或模态上分多个阶段训练模型组件，CoCa 则通过将所有标签都视为文本，直接利用各种数据源（即带标注的图像和带噪声替代文本的图像）从零开始进行端到端预训练，同时用于对比目标和生成目标。

3.3 Contrastive Captioners for Downstream Tasks

Zero-shot Transfer. A pretrained CoCa model performs many tasks in a zero-shot manner by leveraging both image and text inputs, including zero-shot image classification, zero-shot image-text cross-retrieval, zero-shot video-text cross-retrieval. Following previous practices [12, 32], "zero-shot" here is different from classical zero-shot learning in that during pretraining, the model may see relevant supervised information, but no supervised examples are used during the transfer protocol. For the pretraining data, we follow strict de-duplication procedures introduced in [13, 32] to filter all near-domain examples to our downstream tasks.

Frozen-feature Evaluation. As discussed in the previous section, CoCa adopts task-specific attentional pooling [42] (*pooler* for brevity) to customize visual representations for different types downstream tasks while sharing the backbone encoder. This enables the model to obtain strong performance as a *frozen encoder* where we only learn a new pooler to aggregate features. It can also benefit to multi-task problems that share the same frozen image encoder computation but different task-specific heads. As also discussed in [23], linear-evaluation struggles to accurately measure learned representations and we find the attentional poolers are more practical for real-world applications.

CoCa for Video Action Recognition. We use a simple approach to enable a learned CoCa model for video action recognition tasks. We first take multiple frames of a video and feed each frame into the *shared* image encoder individually as shown in Figure 3. For frozen-feature evaluation or finetuning, we learn an additional pooler on top of the spatial and temporal feature tokens with a softmax cross-entropy loss. Note the pooler has a single query token thus the computation of pooling over all spatial and temporal tokens is not expensive. For zero-shot video-text retrieval, we use an even simpler approach by computing the mean embedding of 16 frames of the video (frames are uniformly sampled from a video). We also encode the captions of each video as target embeddings when computing retrieval metrics.

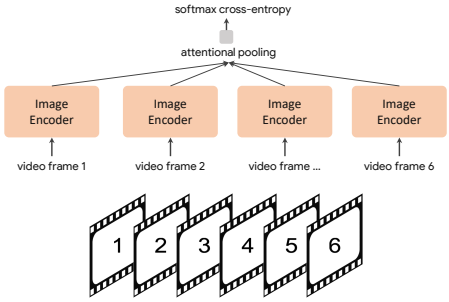


Figure 3: CoCa for video recognition.

4 Experiments

In this section, we first describe the details of our experimental setup. The main results are presented next organized as visual recognition tasks, crossmodal alignment tasks, image captioning and multimodal understanding tasks. Our main results are conducted under three categories for downstream tasks: zero-shot transfer, frozen-feature evaluation and finetuning. We also present ablation experiments including training objectives and architecture designs.

4.1 Training Setup

Data. As discussed in Section 3.2, CoCa is pretrained from scratch in a single stage on both web-scale alt-text data and annotated images by treating all labels simply as texts. We use the JFT-3B dataset [21] with label names as the paired texts, and the ALIGN dataset [13] with noisy alt-texts. Similar to [33], we randomly shuffle and concatenate label names of each image in JFT together with a prompt sampled from [12]. An example of the resulting text label of a JFT image would look like "a photo of the cat, animal". Unlike prior models [32, 33] that also use the combination of these two datasets, we train all model parameters from scratch at the same time without pretraining an image encoder with supervised cross-entropy loss for simplicity and pretraining efficiency. To ensure fair evaluation, we follow the strict de-duplication procedures introduced in [13, 32] to filter all near-domain examples (3.6M images are removed in total) to our downstream tasks. To tokenize text input, we use a sentence-piece model [43, 44] with a vocabulary size of 64k trained on the sampled pretraining dataset.

Optimization. Our models are implemented in the Lingvo framework [45] with GSPMD [46, 47, 48, 49] for scaling performance. Following [33], we use a batch size of 65,536 image-text pairs,

3.3 面向下游任务的对比式描述生成器

零样本迁移。预训练的 CoCa 模型通过同时利用图像和文本输入，以零样本方式执行多种任务，包括零样本图像分类、零样本图像-文本交叉检索、零样本视频-文本交叉检索。遵循以往做法 [12, 32]，这里的“零样本”不同于经典的零样本学习，因为在预训练期间，模型可能会看到相关的监督信息，但在迁移协议中不使用任何监督样本。对于预训练数据，我们遵循 [13, 32] 中提出的严格去重流程，以过滤掉所有与下游任务近域的样本。

冻结特征评估。如上一节所述，CoCa 采用任务特定的注意力池化 [42]（为简洁起见称为 pooler），以在共享骨干编码器的同时，为不同类型的下游任务定制视觉表征。这使得模型能够作为冻结编码器获得强大的性能，此时我们只需学习一个新的 pooler 来聚合特征。它也有利于多任务问题，即共享相同的冻结图像编码器计算，但使用不同的任务特定头。如 [23] 中也讨论的那样，线性评估难以准确衡量学习到的表征，而我们发现注意力池化器在现实应用中更加实用。

用于视频动作识别的 CoCa。我们采用一种简单的方法，使学习到的 CoCa 模型能够用于视频动作识别任务。我们首先从视频中提取多个帧，并将每一帧分别输入共享的图像编码器，如图 3 所示。对于冻结特征评估或微调，我们在空间和时间特征 token 之上额外学习一个池化器，并使用 softmax 交叉熵损失。注意，该池化器只有一个查询 token，因此对所有空间和时间 token 进行池化的计算开销并不大。对于零样本视频-文本检索，我们采用一种更简单的方法，即计算视频中 16 帧的平均嵌入（这些帧从视频中均匀采样）。在计算检索指标时，我们还将每个视频的字幕编码为目标嵌入。

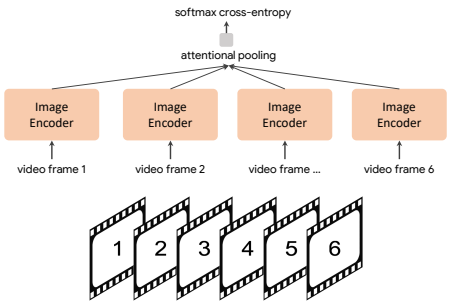


图 3：用于视频识别的 CoCa。

4 实验

在本节中，我们首先描述实验设置的细节。接下来，我们按视觉识别任务、跨模态对齐任务、图像描述生成任务和多模态理解任务来组织并呈现主要结果。我们的主要结果针对下游任务分为三类：零样本迁移、冻结特征评估和微调。我们还展示了包括训练目标和架构设计在内的消融实验。

4.1 训练设置

数据。如第 3.2 节所述，CoCa 通过将所有标签简单地视为文本，在单阶段中从头开始同时在网络规模的替代文本数据和带标注图像上进行预训练。我们使用 JFT-3B 数据集 [21]，将标签名称作为配对文本；并使用 ALIGN 数据集 [13] 及其噪声替代文本。类似于 [33]，我们将 JFT 中每张图像的标签名称随机打乱并拼接，同时加入一个从 [12] 中采样的提示词。由此得到的 JFT 图像文本标签示例如下：“a photo of the cat, animal”。不同于以往同样使用这两个数据集组合的模型 [32, 33]，为简化流程并提高预训练效率，我们同时从头开始训练所有模型参数，而不使用监督交叉熵损失预训练图像编码器。为确保公平评估，我们遵循 [13, 32] 中提出的严格去重流程，过滤掉与下游任务近域的所有样本（总共移除了 360 万张图像）。为对文本输入进行分词，我们使用在采样的预训练数据集上训练的 sentence-piece 模型 [43, 44]，其词表大小为 64k。

优化。我们的模型在 Lingvo 框架 [45] 中实现，并使用 GSPMD [46, 47, 48, 49] 来提升扩展性能。按照 [33]，我们使用 65,536 个图像-文本对的批量大小，

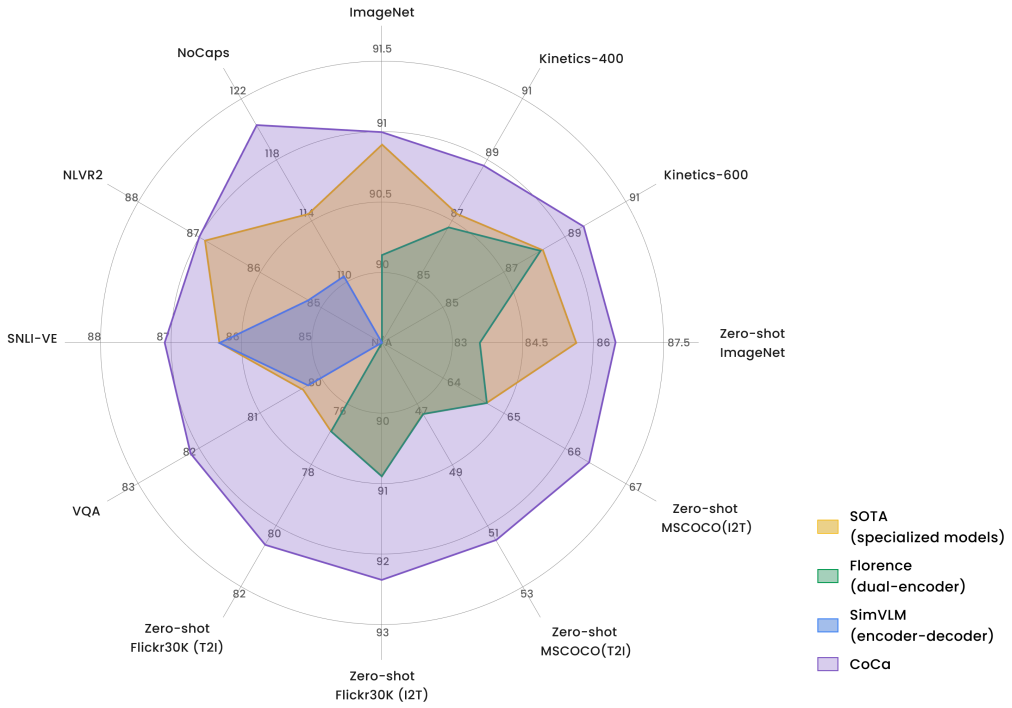


Figure 4: Comparison of CoCa with other image-text foundation models (without task-specific customization) and multiple state-of-the-art task-specialized models.

Model	ImageNet
ALIGN [13]	88.6
Florence [14]	90.1
MetaPseudoLabels [51]	90.2
CoAtNet [10]	90.9
ViT-G [21]	90.5
+ Model Soups [52]	90.9
CoCa (frozen)	90.6
CoCa (finetuned)	**91.0**

Model	K-400	K-600	K-700	Moments-in-Time
ViViT [53]	84.8	84.3	-	38.0
MoViNet [54]	81.5	84.8	79.4	40.2
VATT [55]	82.1	83.6	-	41.1
Florence [14]	86.8	88.0	-	-
MaskFeat [56]	87.0	88.3	80.4	
CoVeR [11]	87.2	87.9	78.5	46.1
CoCa (frozen)	88.0	88.5	81.1	47.4
CoCa (finetuned)	**88.9**	**89.4**	**82.7**	**49.0**

Table 2: Image classification and video action recognition with frozen encoder or finetuned encoder.

where half of each batch comes from JFT and ALIGN, respectively. All models are trained on the combined contrastive and captioning objectives in Eq.(4) for 500k steps, roughly corresponding to 5 epochs on JFT and 10 epochs on ALIGN. As shown later in our studies, we find a larger captioning loss weight is better and thus $\lambda_{\text{Cap}} = 2.0$ and $\lambda_{\text{Con}} = 1.0$. Following [13], we apply a contrastive loss with a trainable temperature τ with an initial value of 0.07. For memory efficiency, we use the Adafactor [50] optimizer with $\beta_1 = 0.9, \beta_2 = 0.999$ and decoupled weight decay ratio of 0.01. We warm up the learning rate for the first 2% of training steps to a peak value of 8×10^{-4}, and linearly decay it afterwards. Pretraining CoCa takes about 5 days on 2,048 CloudTPUv4 chips. Following [12, 13, 14], we continue pretraining for one epoch on a higher resolution of 576×576. For finetuning evaluation, we mainly follow simple protocols and directly train CoCa on downstream tasks without further metric-specific tuning like CIDEr scores (details in Appendix A and B).

4.2 Main Results

We extensively evaluate the capabilities of CoCa models on a wide range of downstream tasks as a pretrained foundation model. We mainly consider core tasks of three categories that examine (1)

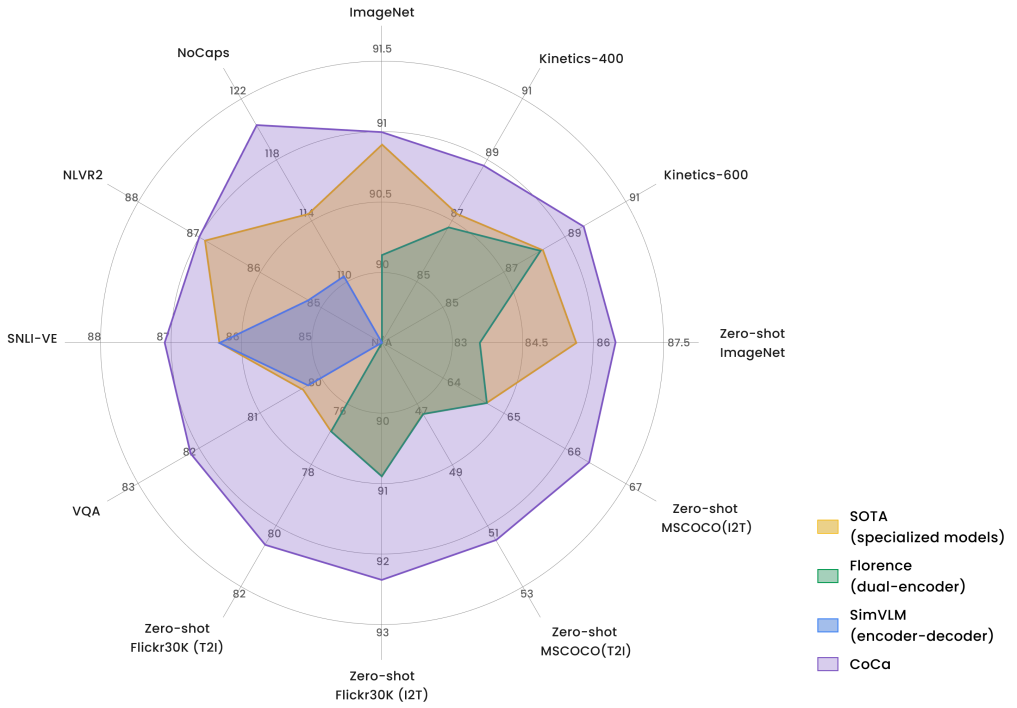


图 4：CoCa 与其他图文基础模型（无任务特定定制）以及多个最先进的任务专用模型的比较。

Model	ImageNet
ALIGN [13]	88.6
Florence [14]	90.1
MetaPseudoLabels [51]	90.2
CoAtNet [10]	90.9
ViT-G [21]	90.5
+ Model Soups [52]	90.9
CoCa (frozen)	90.6
CoCa (finetuned)	**91.0**

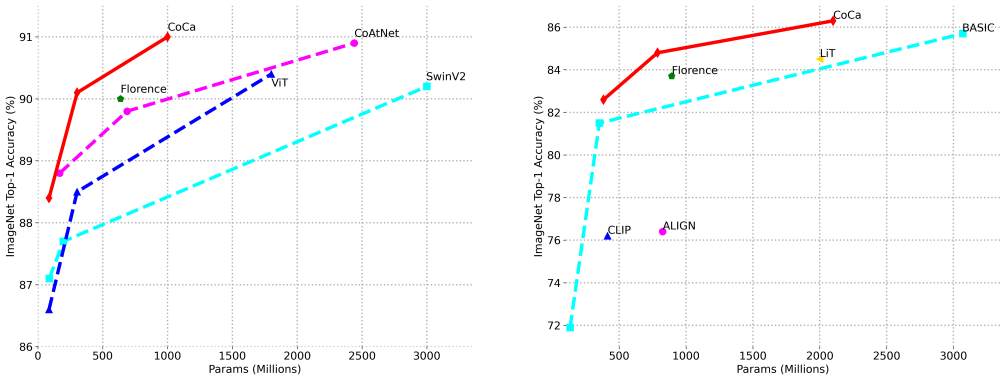
Model	K-400	K-600	K-700	Moments-in-Time
ViViT [53]	84.8	84.3	-	38.0
MoViNet [54]	81.5	84.8	79.4	40.2
VATT [55]	82.1	83.6	-	41.1
Florence [14]	86.8	88.0	-	-
MaskFeat [56]	87.0	88.3	80.4	
CoVeR [11]	87.2	87.9	78.5	46.1
CoCa (frozen)	88.0	88.5	81.1	47.4
CoCa (finetuned)	**88.9**	**89.4**	**82.7**	**49.0**

表 2：使用冻结编码器或微调编码器进行图像分类和视频动作识别。

其中，每个批次的一半分别来自 JFT 和 ALIGN。所有模型都基于式(4)中的组合式对比学习和字幕生成目标训练 500k 步，大致相当于在 JFT 上训练 5 个 epoch、在 ALIGN 上训练 10 个 epoch。正如我们后续研究所示，我们发现更大的字幕生成损失权重效果更好，因此 $\lambda_{\text{Cap}} = 2.0$ 且 $\lambda_{\text{Con}} = 1.0$。遵循 [13]，我们采用带有可训练温度 τ 的对比损失，其初始值为 0.07。为提高内存效率，我们使用 Adafactor [50] 优化器，其中 $\beta_1 = 0.9, \beta_2 = 0.999$，并采用 0.01 的解耦权重衰减比例。我们在训练前 2% 的步数中将学习率预热到峰值 8×10^{-4}，随后进行线性衰减。CoCa 的预训练在 2,048 个 CloudTPUv4 芯片上大约需要 5 天。遵循 [12, 13, 14]，我们继续在更高分辨率 576×576 上预训练一个 epoch。对于微调评估，我们主要遵循简单协议，直接在下游任务上训练 CoCa，而不进行如 CIDEr 分数等进一步的指标特定调优（详见附录 A 和 B）。

4.2 主要结果

我们广泛评估了 CoCa 模型作为预训练基础模型在各种下游任务中的能力。我们主要考虑三类核心任务，以考察（1）



(a) Finetuned ImageNet Top-1 Accuracy.

(b) Zero-Shot ImageNet Top-1 Accuracy.

Figure 5: Image classification scaling performance of model sizes.

visual recognition, (2) crossmodal alignment, and (3) image captioning and multimodal understanding capabilities. Since CoCa produces both aligned unimodal representations and fused multimodal embeddings at the same time, it is easily transferable to all three task groups with minimal adaption. Figure 4 summarizes the performance on key benchmarks of CoCa compared to other dual-encoder and encoder-decoder foundation models and state-of-the-art task-specialized methods. CoCa sets new state-of-the-art results on tasks of all three categories with a single pretrained checkpoint.

4.2.1 Visual Recognition Tasks

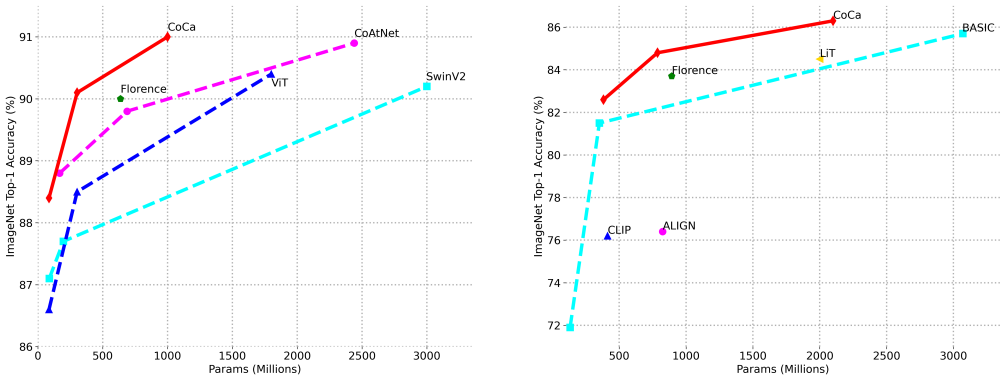
Our visual recognition experiments are conducted on ImageNet [9] as image recognition benchmark, and multiple video datasets including Kinetics-400 [57], Kinetics-600 [58], Kinetics-700 [59], Moments-in-Time [60] as test-beds for video action recognition; it is noteworthy that CoCa pretrains on image data only, without accessing any extra video datasets. We apply the CoCa encoder on video frames individually (Section 3.3) without early fusion of temporal information, yet the resulting CoCa-for-Video model performs better than many spatio-temporal early-fused video models.

Frozen-feature. We apply a pretrained frozen CoCa model on both image classification and video action recognition. The encoder is used for both tasks while the decoder is discarded. As discussed in Section 3.3, an attentional pooling is learned together with a softmax cross-entropy loss layer on top of the embedding outputs from CoCa encoder. For video classification, a single query-token is learned to weight outputs of all tokens of spatial patches $\times$ temporal frames. We set a learning rate of 5×10^{-4} on both attentional pooler and softmax, batch size of 128, and a cosine learning rate schedule (details in Appendix A). For video action recognition, we compare CoCa with other approaches on the same setup (*i.e.*, without extra supervised video data and without audio signals as model inputs). As shown in Table 2, without finetuning full encoder, CoCa already achieves competitive Top-1 classification accuracies compared to specialized image and outperforms prior state-of-the-art specialized methods on video tasks.

Finetuning. Based on the architecture of frozen-feature evaluation, we further finetune CoCa encoders on image and video datasets individually with a smaller learning rate of 1×10^{-4}. More experimental details are summarized in the Appendix A. The finetuned CoCa has improved performance across these tasks. Notably, CoCa obtains new state-of-the-art 91.0% Top-1 accuracy on ImageNet, as well as better video action recognition results compared with recent video approaches. More importantly, CoCa models use much less parameters than other methods in the visual encoder as shown in Figure 5a. These results suggest the proposed framework efficiently combines text training signals and thus is able to learn high-quality visual representation better than the classical single-encoder approach.

4.2.2 Crossmodal Alignment Tasks

Unlike other fusion-based foundation methods [16, 17, 35], CoCa is naturally applicable to crossmodal alignment tasks since it generates aligned image and text unimodal embeddings. In particular, we



(a) Finetuned ImageNet Top-1 Accuracy.

(b) Zero-Shot ImageNet Top-1 Accuracy.

图 5：不同模型规模的图像分类扩展性能。

视觉识别、(2) 跨模态对齐，以及 (3) 图像字幕生成和多模态理解能力。由于 CoCa 能够同时生成对齐的单模态表示和融合的多模态嵌入，因此只需极少的适配即可轻松迁移到这三类任务。图 4 总结了 CoCa 与其他双编码器和编码器-解码器基础模型以及最先进的任务专用方法在关键基准上的性能对比。CoCa 通过单个预训练检查点，在所有三类任务上都取得了新的最先进结果。

4.2.1 视觉识别任务

我们的视觉识别实验在 ImageNet [9] 上进行，将其作为图像识别基准；同时使用多个视频数据集，包括 Kinetics-400 [57]、Kinetics-600 [58]、Kinetics-700 [59]、Moments-in-Time [60]，作为视频动作识别的测试平台。值得注意的是，CoCa 仅在图像数据上进行预训练，并未访问任何额外的视频数据集。我们将 CoCa 编码器分别应用于视频帧（第 3.3 节），没有对时间信息进行早期融合，但由此得到的 CoCa-for-Video 模型表现优于许多进行时空早期融合的视频模型。

冻结特征。我们将一个预训练的冻结 CoCa 模型应用于图像分类和视频动作识别。编码器用于这两项任务，而解码器被丢弃。如第 3.3 节所述，在 CoCa 编码器的嵌入输出之上，同时学习一个注意力池化层和一个 softmax 交叉熵损失层。对于视频分类，学习一个单一的查询 token，用于对所有空间 patch $\times$ 时间帧的 token 输出进行加权。我们将注意力池化器和 softmax 的学习率均设为 5×10^{-4}，批量大小设为 128，并采用余弦学习率调度（详见附录 A）。对于视频动作识别，我们在相同设置下将 CoCa 与其他方法进行比较（即不使用额外的有监督视频数据，也不将音频信号作为模型输入）。如表 2 所示，在不微调整个编码器的情况下，CoCa 已经达到了可与专门的图像方法相媲美的 Top-1 分类准确率，并且在视频任务上优于此前最先进的专门方法。

微调。基于冻结特征评估的架构，我们进一步分别在图像和视频数据集上以较小的学习率 1×10^{-4} 对 CoCa 编码器进行微调。更多实验细节汇总于附录 A。经过微调的 CoCa 在这些任务上的性能均有所提升。值得注意的是，CoCa 在 ImageNet 上取得了新的最先进水平，即 91.0% 的 Top-1 准确率；同时，与近期的视频方法相比，也获得了更好的视频动作识别结果。更重要的是，如图 5a 所示，CoCa 模型在视觉编码器中使用的参数量远少于其他方法。这些结果表明，所提出的框架能够高效地结合文本训练信号，因此相比经典的单编码器方法，能够更好地学习高质量的视觉表示。

4.2.2 跨模态对齐任务

与其他基于融合的基础方法 [16, 17, 35] 不同，CoCa 天然适用于跨模态对齐任务，因为它会生成对齐的图像和文本单模态嵌入。具体而言，我们

	Flickr30K (1K test set)						MSCOCO (5K test set)					
	Image → Text			Text → Image			Image → Text			Text → Image		
Model	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
CLIP [12]	88.0	98.7	99.4	68.7	90.6	95.2	58.4	81.5	88.1	37.8	62.4	72.2
ALIGN [13]	88.6	98.7	99.7	75.7	93.8	96.8	58.6	83.0	89.7	45.6	69.8	78.6
FLAVA [35]	67.7	94.0	-	65.2	89.4	-	42.7	76.8	-	38.4	67.5	-
FILIP [61]	89.8	99.2	99.8	75.0	93.4	96.3	61.3	84.3	90.4	45.9	70.6	79.3
Florence [14]	90.9	99.1	-	76.7	93.6	-	64.7	85.9	-	47.2	71.4	-
CoCa-Base	89.8	98.8	99.8	76.8	93.7	96.8	63.8	84.7	90.7	47.5	72.4	80.9
CoCa-Large	91.4	99.2	99.9	79.0	95.1	97.4	65.4	85.6	91.4	50.1	73.8	81.8
CoCa	**92.5**	**99.5**	**99.9**	**80.4**	**95.7**	**97.7**	**66.3**	**86.2**	**91.8**	**51.2**	**74.2**	**82.0**

Table 3: Zero-shot image-text retrieval results on Flickr30K [62] and MSCOCO [63] datasets.

Model	ImageNet	ImageNet-A	ImageNet-R	ImageNet-V2	ImageNet-Sketch	ObjectNet	Average
CLIP [12]	76.2	77.2	88.9	70.1	60.2	72.3	74.3
ALIGN [13]	76.4	75.8	92.2	70.1	64.8	72.2	74.5
FILIP [61]	78.3	-	-	-	-	-	-
Florence [14]	83.7	-	-	-	-	-	-
LiT [32]	84.5	79.4	93.9	78.7	-	81.1	-
BASIC [33]	85.7	85.6	95.7	80.6	76.1	78.9	83.7
CoCa-Base	82.6	76.4	93.2	76.5	71.7	71.6	78.7
CoCa-Large	84.8	85.7	95.6	79.6	75.7	78.6	83.3
CoCa	**86.3**	**90.2**	**96.5**	**80.7**	**77.6**	**82.7**	**85.7**

Table 4: Zero-shot image classification results on ImageNet [9], ImageNet-A [64], ImageNet-R [65], ImageNet-V2 [66], ImageNet-Sketch [67] and ObjectNet [68].

are interested in the zero-shot setting where all parameters are frozen after pretraining and directly used to extract embeddings. Here, we use the same embeddings used for contrastive loss during pretraining, and thus the multimodal text decoder is not used.

Zero-Shot Image-Text Retrieval. We evaluate CoCa on the two standard image-text retrieval benchmarks: MSCOCO [63] and Flickr30K [62]. Following the CLIP setting [12], we first independently feed each image/text to the corresponding encoder and obtain embeddings for all image/text in the test set. We then retrieve based on cosine similarity scores over the whole test set. As shown in Table 3, CoCa significantly improves over prior methods on both image-to-text and text-to-image retrievals on all metrics. In addition, our model is parameter-efficient, with CoCa-Base already outperforming strong baselines (CLIP [12] and ALIGN [13]) and CoCa-Large outperforming Florence [14] (which contains a parameter count comparable to ViT-Huge). This shows that CoCa learns good unimodal representations *and* aligns them well across modalities.

Zero-Shot Image Classification. Following prior work [12, 13], we use the aligned image/text embeddings to perform zero-shot image classification by matching images with label names without finetuning. We follow the exact setup in [12] and apply the same set of prompts used for label class names. As shown in Table 4, CoCa sets new state-of-the-art zero-shot classification results on ImageNet. Notably, CoCa uses fewer parameters than prior best model [33] while smaller CoCa variants already outperform strong baselines [12, 14], as shown in Figure 5b. In addition, our model demonstrates effective generalization under zero-shot evaluation, consistent with prior findings [12, 13], with CoCa improving on all six datasets considered. Lastly, while prior models [32, 33] found sequentially pretraining with single-encoder and dual-encoder methods in multiple stages is crucial to performance gains, our results show it is possible to attain strong performance by unifying training objectives and datasets in a single-stage framework.

Zero-Shot Video Retrieval. We evaluate video-text retrieval using CoCa on MSR-VTT [71] using the full split. Table 5 shows that CoCa produces the highest retrieval metrics for both text-to-video and video-to-text retrieval. It is important to note that MSR-VTT videos are sourced from YouTube, and we require the original videos to compute our embeddings. Many of the videos have been made explicitly unavailable [72], hence we compute retrieval over the subset of data that is publicly available at the time of evaluation. Using code[3] provided by the authors of Socratic Models [70], we re-computed metrics on the available subset for those methods, indicated by "(subset)" for fairest comparison.

[3] https://socraticmodels.github.io

	Flickr30K (1K test set)						MSCOCO (5K test set)					
	Image → Text			Text → Image			Image → Text			Text → Image		
Model	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
CLIP [12]	88.0	98.7	99.4	68.7	90.6	95.2	58.4	81.5	88.1	37.8	62.4	72.2
ALIGN [13]	88.6	98.7	99.7	75.7	93.8	96.8	58.6	83.0	89.7	45.6	69.8	78.6
FLAVA [35]	67.7	94.0	-	65.2	89.4	-	42.7	76.8	-	38.4	67.5	-
FILIP [61]	89.8	99.2	99.8	75.0	93.4	96.3	61.3	84.3	90.4	45.9	70.6	79.3
Florence [14]	90.9	99.1	-	76.7	93.6	-	64.7	85.9	-	47.2	71.4	-
CoCa-Base	89.8	98.8	99.8	76.8	93.7	96.8	63.8	84.7	90.7	47.5	72.4	80.9
CoCa-Large	91.4	99.2	99.9	79.0	95.1	97.4	65.4	85.6	91.4	50.1	73.8	81.8
CoCa	**92.5**	**99.5**	**99.9**	**80.4**	**95.7**	**97.7**	**66.3**	**86.2**	**91.8**	**51.2**	**74.2**	**82.0**

表3：在 Flickr30K [62] 和 MSCOCO [63] 数据集上的零样本图像-文本检索结果。

Model	ImageNet	ImageNet-A	ImageNet-R	ImageNet-V2	ImageNet-Sketch	ObjectNet	Average
CLIP [12]	76.2	77.2	88.9	70.1	60.2	72.3	74.3
ALIGN [13]	76.4	75.8	92.2	70.1	64.8	72.2	74.5
FILIP [61]	78.3	-	-	-	-	-	-
Florence [14]	83.7	-	-	-	-	-	-
LiT [32]	84.5	79.4	93.9	78.7	-	81.1	-
BASIC [33]	85.7	85.6	95.7	80.6	76.1	78.9	83.7
CoCa-Base	82.6	76.4	93.2	76.5	71.7	71.6	78.7
CoCa-Large	84.8	85.7	95.6	79.6	75.7	78.6	83.3
CoCa	**86.3**	**90.2**	**96.5**	**80.7**	**77.6**	**82.7**	**85.7**

表 4：在 ImageNet [9]、ImageNet-A [64]、ImageNet-R [65]、ImageNet-V2 [66]、ImageNet-Sketch [67] 和 ObjectNet [68] 上的零样本图像分类结果。

我们关注零样本设置，其中所有参数在预训练后被冻结，并直接用于提取嵌入。这里，我们使用与预训练期间用于对比损失相同的嵌入，因此不使用多模态文本解码器。

零样本图文检索。我们在两个标准图文检索基准上评估 CoCa：MSCOCO [63] 和 Flickr30K [62]。按照 CLIP 的设置 [12]，我们首先将每张图像/每段文本分别输入相应的编码器，并获得测试集中所有图像/文本的嵌入表示。然后，我们基于整个测试集上的余弦相似度分数进行检索。如表 3 所示，CoCa 在所有指标上，在图像到文本和文本到图像检索任务中都显著优于以往方法。此外，我们的模型具有参数效率，CoCa-Base 已经超过了强基线（CLIP [12] 和 ALIGN [13]），而 CoCa-Large 超过了 Florence [14]（其参数量与 ViT-Huge 相当）。这表明 CoCa 学习到了良好的单模态表示，并能在不同模态之间很好地对齐它们。

零样本图像分类。遵循先前工作 [12, 13]，我们使用对齐的图像/文本嵌入，通过将图像与标签名称进行匹配，在不进行微调的情况下执行零样本图像分类。我们遵循 [12] 中完全相同的设置，并应用同一组用于标签类别名称的提示。如表 4 所示，CoCa 在 ImageNet 上取得了新的最先进零样本分类结果。值得注意的是，与此前最佳模型 [33] 相比，CoCa 使用的参数更少，而较小的 CoCa 变体已经优于强基线 [12, 14]，如图 5b 所示。此外，我们的模型在零样本评估下表现出有效的泛化能力，这与先前发现 [12, 13] 一致，CoCa 在所考虑的全部六个数据集上均有所提升。最后，尽管先前模型 [32, 33] 发现，以多个阶段依次采用单编码器和双编码器方法进行预训练对于性能提升至关重要，但我们的结果表明，通过在单阶段框架中统一训练目标和数据集，也可以获得强大的性能。

零样本视频检索。我们在 MSR-VTT [71] 的完整划分上使用 CoCa 评估视频-文本检索。表 5 显示，CoCa 在文本到视频和视频到文本检索中均取得了最高的检索指标。需要注意的是，MSR-VTT 视频来源于 YouTube，而我们需要原始视频来计算嵌入。许多视频已被明确设为不可用 [72]，因此我们在评估时基于公开可用的数据子集计算检索结果。使用 Socratic Models [70] 作者提供的 code[3]，我们在可用子集上重新计算了这些方法的指标，并以“(subset)”标注，以便进行最公平的比较。

[3] `https://socraticmodels.github.io`

Method	MSR-VTT Full					
	Text → Video			Video → Text		
	R@1	R@5	R@10	R@1	R@5	R@10
CLIP [69]	21.4	41.1	50.4	40.3	69.7	79.2
Socratic Models [70]	-	-	-	44.7	71.2	80.0
CLIP [69] (subset)	23.3	44.2	53.6	43.3	73.3	**81.8**
Socratic Models [70] (subset)	-	-	-	46.9	**73.5**	81.3
CoCa (subset)	**30.0**	**52.4**	**61.6**	**49.9**	73.4	81.4

Table 5: Zero-shot Video-Text Retrieval on MSR-VTT Full test set.

Model	VQA		SNLI-VE		NLVR2	
	test-dev	test-std	dev	test	dev	test-p
UNITER [26]	73.8	74.0	79.4	79.4	79.1	80.0
VinVL [27]	76.6	76.6	-	-	82.7	84.0
CLIP-ViL [73]	76.5	76.7	80.6	80.2	-	-
ALBEF [36]	75.8	76.0	80.8	80.9	82.6	83.1
BLIP [37]	78.3	78.3	-	-	82.2	82.2
OFA [17]	79.9	80.0	90.3†	90.2†	-	-
VLMo [30]	79.9	80.0	-	-	85.6	86.9
SimVLM [16]	80.0	80.3	86.2	86.3	84.5	85.2
Florence [14]	80.2	80.4	-	-	-	-
METER [74]	80.3	80.5	-	-	-	-
CoCa	**82.3**	**82.3**	**87.0**	**87.1**	**86.1**	**87.0**

Table 6: Multimodal understanding results comparing vision-language pretraining methods. †OFA uses both image and text premises as inputs while other models utilize the image only.

4.2.3 Image Captioning and Multimodal Understanding Tasks

Another key advantage of CoCa is its ability to process multimodal embeddings as an encoder-decoder model trained with the generative objective. Therefore, CoCa can perform both image captioning and multimodal understanding downstream tasks without any further fusion adaptation [73, 74]. Overall, experimental results suggest CoCa reaps the benefit of a encoder-decoder model to obtain strong multimodal understanding and generation capabilities, in addition to the vision and retrieval capabilities as a dual-encoder method.

Multimodal Understanding. As shown in [16], the output of encoder-decoder models can jointly encode image and text inputs, and can be used for tasks that require reasoning over both modalities. We consider three popular multimodal understaning benchmarks: visual question answering (VQA v2 [75]), visual entailment (SNLI-VE [76]), and visual reasoning (NLVR2 [77]). We mainly follow the settings in [16] and train linear classifiers on top of the decoder outputs to predict answers (more details in Appendix B). Our results in Table 6 suggest that CoCa outperforms strong vision-language pretraining (VLP) baselines and obtains state-of-the-art performance on all three tasks. While prior dual-encoder models [12, 14] do not contain fusion layers and thus require an additional VL pretraining stage for downstream multimodal understanding tasks, CoCa subsumes the three pretraining paradigms and obtains better performance on VL tasks with lightweight finetuning.

Image Captioning. In addition to multimodal classification tasks, CoCa is also directly applicable to image captioning tasks as an encoder-decoder model. We finetune CoCa with the captioning loss $\mathcal{L}_{\text{Cap}}$ only on MSCOCO [63] captioning task and evaluate on both MSCOCO Karpathy-test split and NoCaps [78] online evaluation. As shown by experiments in Table 7, CoCa outperforms strong baselines trained with cross-entropy loss on MSCOCO, and achieves results comparable to methods with CIDEr metric-specific optimization [79]. It is noteworthy that we do not use CIDEr-specific optimization [79] for simplicity. On the challenging NoCaps benchmark, CoCa obtains new state-of-the-art on both validation and test splits (generated examples shown in Figure 6). These results showcase the generative capability of CoCa as an image-text foundation model.

	MSR-VTT Full					
	Text → Video			Video → Text		
Method	R@1	R@5	R@10	R@1	R@5	R@10
CLIP [69]	21.4	41.1	50.4	40.3	69.7	79.2
Socratic Models [70]	-	-	-	44.7	71.2	80.0
CLIP [69] (subset)	23.3	44.2	53.6	43.3	73.3	**81.8**
Socratic Models [70] (subset)	-	-	-	46.9	**73.5**	81.3
CoCa (subset)	**30.0**	**52.4**	**61.6**	**49.9**	73.4	81.4

Table 5: Zero-shot Video-Text Retrieval on MSR-VTT Full test set.

	VQA		SNLI-VE		NLVR2	
Model	test-dev	test-std	dev	test	dev	test-p
UNITER [26]	73.8	74.0	79.4	79.4	79.1	80.0
VinVL [27]	76.6	76.6	-	-	82.7	84.0
CLIP-ViL [73]	76.5	76.7	80.6	80.2	-	-
ALBEF [36]	75.8	76.0	80.8	80.9	82.6	83.1
BLIP [37]	78.3	78.3	-	-	82.2	82.2
OFA [17]	79.9	80.0	90.3†	90.2†	-	-
VLMo [30]	79.9	80.0	-	-	85.6	86.9
SimVLM [16]	80.0	80.3	86.2	86.3	84.5	85.2
Florence [14]	80.2	80.4	-	-	-	-
METER [74]	80.3	80.5	-	-	-	-
CoCa	**82.3**	**82.3**	**87.0**	**87.1**	**86.1**	**87.0**

表 6：多模态理解结果，对比视觉-语言预训练方法。†OFA 同时使用图像和文本前提作为输入，而其他模型仅使用图像。

4.2.3 图像描述与多模态理解任务

CoCa 的另一个关键优势在于，作为采用生成式目标训练的编码器-解码器模型，它能够处理多模态嵌入。因此，CoCa 无需任何进一步的融合适配，就可以执行图像字幕生成和多模态理解等下游任务 [73, 74]。总体而言，实验结果表明，CoCa 既能作为双编码器方法具备视觉和检索能力，又能受益于编码器-解码器模型，从而获得强大的多模态理解与生成能力。

多模态理解。如 [16] 所示，编码器-解码器模型的输出可以联合编码图像和文本输入，并可用于需要对两种模态进行推理的任务。我们考虑了三个流行的多模态理解基准：视觉问答（VQA v2 [75]）、视觉蕴含（SNLI-VE [76]）和视觉推理（NLVR2 [77]）。我们主要遵循 [16] 中的设置，并在解码器输出之上训练线性分类器来预测答案（更多细节见附录 B）。表 6 中的结果表明，CoCa 优于强大的视觉-语言预训练（VLP）基线，并在所有三个任务上取得了最先进的性能。以往的双编码器模型 [12, 14] 不包含融合层，因此需要额外的视觉-语言（VL）预训练阶段来完成下游多模态理解任务；而 CoCa 涵盖了这三种预训练范式，并通过轻量级微调在视觉-语言任务上获得了更好的性能。

图像描述生成。除了多模态分类任务外，CoCa 作为编码器-解码器模型也可直接应用于图像描述生成任务。我们仅使用描述生成损失 $\mathcal{L}_{\text{Cap}}$ 在 MSCOCO [63] 描述生成任务上对 CoCa 进行微调，并在 MSCOCO Karpathy 测试划分和 NoCaps [78] 在线评测上进行评估。如表 7 的实验所示，CoCa 优于在 MSCOCO 上使用交叉熵损失训练的强基线模型，并取得了与采用 CIDEr 指标特定优化 [79] 的方法相当的结果。值得注意的是，为简化起见，我们没有使用 CIDEr 特定优化 [79]。在具有挑战性的 NoCaps 基准上，CoCa 在验证集和测试集划分上均取得了新的最先进结果（生成示例如图 6 所示）。这些结果展示了 CoCa 作为图像-文本基础模型的生成能力。

	MSCOCO				NoCaps			
					Valid		Test	
	B@4	M	C	S	C	S	C	S
CLIP-ViL [73]	40.2	29.7	134.2	23.8	-	-	-	-
BLIP [37]	40.4	-	136.7	-	113.2	14.8	-	-
VinVL[27]	41.0	31.1	140.9	25.4	105.1	14.4	103.7	14.4
SimVLM [16]	40.6	33.7	143.3	**25.4**	112.2	-	110.3	14.5
LEMON [80]	**41.5**	30.8	139.1	24.1	117.3	15.0	114.3	14.9
LEMON$_{\text{SCST}}$ [80]†	42.6	31.4	145.5	25.5	-	-	-	-
OFA [17]†	43.5	31.9	149.6	26.1	-	-	-	-
CoCa	40.9	**33.9**	**143.6**	24.7	**122.4**	**15.5**	**120.6**	**15.5**

Table 7: Image captioning results on MSCOCO and NoCaps (B@4: BLEU@4, M: METEOR, C: CIDEr, S: SPICE). †Models finetuned with CIDEr optimization.



Figure 6: Curated samples of text captions generated by CoCa with NoCaps images as input.

4.3 Ablation Analysis

We extensively ablate the properties of CoCa on a smaller model variant. Specifically, we train CoCa-Base with a reduced 12 decoder layers and a total batch size of 4,096. We mainly evaluate using zero-shot image classification and VQA, since the former covers both visual representation quality and crossmodal alignment, while the later is representative for multimodal reasoning.

Captioning *vs*. Classification. We first examine the effectiveness of captioning loss on image annotation datasets. To do this, we train a naive encoder-decoder model using $\mathcal{L}_{\text{Cap}}$ on the JFT-3B dataset, and compare with a standard ViT-Base single-encoder model trained with $\mathcal{L}_{\text{Cls}}$ in Table 8a. We find encoder-decoder models to perform on par with single-encoder pretraining on both linear evaluation and finetuned results. This suggests that the generative pretraining subsumes classification pretraining, consistent with our intuition that $\mathcal{L}_{\text{Cls}}$ is a special case of $\mathcal{L}_{\text{Cap}}$ when text vocabulary is the set of all possible class names. Thus, our CoCa model can be interpreted as an effective unification of the three paradigms. This explains why CoCa does not need a pretrained visual encoder to perform well.

Training Objectives. We study the effects of the two training objectives and compare CoCa with single-objective variants in Table 8b. Compared to the contrastive-only model, CoCa significantly improves both zero-shot alignment and VQA (notice that the contrastive-only model requires additional fusion for VQA). CoCa performs on par with the captioning-only model on VQA while it additionally enables retrieval-style tasks such as zero-shot classification. Table 8c further studies loss ratios and suggests that the captioning loss not only improves VQA but also zero-shot alignment between modalities. We hypothesize that generative objectives learn fine-grained text representations that further improve text understanding. Finally, we compare training costs in Table 8b (measured in TPUv3-core-days; larger is slower) and find CoCa to be as efficient as the captioning-only model (*a.k.a.*naive encoder-decoder) due to the sharing of compute between two objectives. These suggest combining the two losses induces new capabilities and better performance with minimal extra cost.

Unimodal and Multimodal Decoders. CoCa introduces a novel decoder design and we ablate its components. In Table 8d, we vary the number of unimodal decoder layers (while keeping the total number of layers the same). Intuitively, fewer unimodal text layers leads to worse zero-shot classification due to lack of capacity for good unimodal text understanding, while fewer multimodal layers reduces the model's power to reason over multimodal inputs such as VQA. Overall, we

	MSCOCO				NoCaps			
					Valid		Test	
	B@4	M	C	S	C	S	C	S
CLIP-ViL [73]	40.2	29.7	134.2	23.8	-	-	-	-
BLIP [37]	40.4	-	136.7	-	113.2	14.8	-	-
VinVL[27]	41.0	31.1	140.9	25.4	105.1	14.4	103.7	14.4
SimVLM [16]	40.6	33.7	143.3	**25.4**	112.2	-	110.3	14.5
LEMON [80]	**41.5**	30.8	139.1	24.1	117.3	15.0	114.3	14.9
$\text{LEMON}_{\text{SCST}}$ [80]†	42.6	31.4	145.5	25.5	-	-	-	-
OFA [17]†	43.5	31.9	149.6	26.1	-	-	-	-
CoCa	40.9	**33.9**	**143.6**	24.7	**122.4**	**15.5**	**120.6**	**15.5**

表 7：在 MSCOCO 和 NoCaps 上的图像描述生成结果（B@4：BLEU@4，M：METEOR，C：CIDEr，S：SPICE）。†表示使用 CIDEr 优化进行微调的模型。



图 6：以 NoCaps 图像作为输入，由 CoCa 生成的文本描述精选样本。

4.3 消融分析

我们在一个较小的模型变体上广泛消融分析了 CoCa 的各项属性。具体而言，我们训练了 CoCa-Base，其解码器层数减少为 12 层，总批大小为 4,096。我们主要使用零样本图像分类和 VQA 进行评估，因为前者同时涵盖视觉表征质量和跨模态对齐，而后者则代表多模态推理。

图像描述 vs. 分类。我们首先考察图像描述损失在图像标注数据集上的有效性。为此，我们在 JFT-3B 数据集上使用 $\mathcal{L}_{\text{Cap}}$ 训练了一个朴素的编码器-解码器模型，并在表 8a 中将其与使用 $\mathcal{L}_{\text{Cls}}$ 训练的标准 ViT-Base 单编码器模型进行比较。我们发现，在线性评估和微调结果上，编码器-解码器模型的表现都与单编码器预训练相当。这表明生成式预训练涵盖了分类预训练，这与我们的直觉一致：当文本词汇表是所有可能类别名称的集合时，$\mathcal{L}_{\text{Cls}}$ 是 $\mathcal{L}_{\text{Cap}}$ 的一个特例。因此，我们的 CoCa 模型可以被解释为对这三种范式的有效统一。这解释了为什么 CoCa 不需要预训练的视觉编码器也能表现良好。

训练目标。我们研究了两种训练目标的影响，并在表 8b 中将 CoCa 与单目标变体进行了比较。与仅使用对比学习的模型相比，CoCa 显著提升了零样本对齐和 VQA（注意，仅对比学习模型需要额外的融合模块才能用于 VQA）。CoCa 在 VQA 上的表现与仅使用图像描述生成的模型相当，同时还额外支持零样本分类等检索式任务。表 8c 进一步研究了损失比例，并表明图像描述生成损失不仅提升了 VQA，还改善了模态之间的零样本对齐。我们假设，生成式目标能够学习细粒度的文本表示，从而进一步提升文本理解能力。最后，我们在表 8b 中比较了训练成本（以 TPUv3-core-days 衡量；数值越大表示越慢），发现由于两个目标共享计算，CoCa 与仅使用图像描述生成的模型（即朴素的编码器-解码器）一样高效。这些结果表明，将两种损失结合起来能够以极小的额外成本带来新的能力和更好的性能。

单模态和多模态解码器。CoCa 引入了一种新颖的解码器设计，我们对其组件进行了消融研究。在表 8d 中，我们改变单模态解码器层的数量（同时保持总层数不变）。直观地说，较少的单模态文本层会导致零样本分类效果变差，因为缺乏良好的单模态文本理解能力；而较少的多模态层会降低模型对 VQA 等多模态输入进行推理的能力。总体而言，我们

loss	LE	FT
$\mathcal{L}_{\text{Cls}}$	81.0	85.1
$\mathcal{L}_{\text{Cap}}$	82.1	84.9

(a) Encoder-decoder *vs.* single-encoder models (trained on JFT).

loss	ZS	VQA	TPU cost
$\mathcal{L}_{\text{Con}}$	70.7	59.2	1×
$\mathcal{L}_{\text{Cap}}$	-	68.9	1.17×
$\mathcal{L}_{\textbf{CoCa}}$	**71.6**	**69.0**	**1.18**×

(b) Training objectives ablation.

$\lambda_{\text{Cap}} : \lambda_{\text{Con}}$	ZS	VQA
1:1	71.5	68.6
1:2	71.0	68.1
2:1	**71.6**	**69.0**

(c) Training objectives weights.

n_{uni}	ZS	VQA
3	70.2	69.0
6	**71.6**	**69.0**
9	71.4	68.8

(d) Unimodal decoder layers.

variant	AE	MSCOCO
1 [CLS]	**80.7**	**41.4**
+ text tokens	80.3	40.2
8 [CLS]	80.3	36.9
+ text tokens	80.4	40.3

(e) Contrastive text embedding design ablation.

variant	ZS	VQA
parallel	71.2	68.7
cascade	**71.6**	**69.0**
$n_{\text{query}} = 0$	71.5	69.0
$n_{\text{query}} = 1$	69.3	64.4
$n_{\text{query}} = 32$	71.2	68.2

(f) Attentional pooler design ablation.

Table 8: CoCa ablation experiments. On ImageNet classification, we report top-1 accuracy for: zero-shot (ZS), linear evaluation (LE), attentional evaluation (AE) using pooler on frozen feature, and finetuning (FT). On MSCOCO retrieval, we report the average of image-to-text and text-to-image R@1. On VQA, we report the dev-set vqa score. The default CoCa setting is **bold**.

find decoupling the decoder in half maintains a good balance. One possibility is that global text representation for retrieval doesn't require deep modules [33] while early fusion for shallow layers may also be unnecessary for multimodal understanding. Next, we explore various options to extract unimodal text embeddings. In particular, we experiment with the number of learnable `[CLS]` tokens as well as the aggregation design. For the later, we aggregate over either the `[CLS]` tokens only or the concatenation of `[CLS]` and the original input sentence. Interestingly, in Table 8e we find training a single `[CLS]` token without the original input is preferred for both vision-only and crossmodal retrieval tasks. This indicates that learning an additional simple sentence representation mitigates interference between contrastive and captioning loss, and is powerful enough for strong generalization.

Attentional Poolers. CoCa exploits attentional poolers in its design both for different pretraining objectives and objective-specific downstream task adaptations. In pretraining, we compare a few design variants on using poolers for contrastive loss and generative loss: (1) the "parallel" design which extracts both contrastive and generative losses at the same time on Vision Transformer encoder outputs as shown in Figure 2, and (2) the "cascade" design which applies the contrastive pooler on top of the outputs of the generative pooler. Table 8f shows the results of these variants. Empirically, we find at small scale the "cascade" version (contrastive pooler on top of the generative pooler) performs better and is used by default in all CoCa models. We also study the effect of number of queries where $n_{\text{query}} = 0$ means no generative pooler is used (thus all ViT output tokens are used for decoder cross-attention). Results show that both tasks prefer longer sequences of detailed image tokens at a cost of slightly more computation and parameters. As a result, we use a generative pooler of length 256 to improve multimodal understanding benchmarks while still maintaining the strong frozen-feature capability.

5 Broader Impacts

This work presents an image-text pretraining approach on web-scale datasets that is capable of transferring to a wide range of downstream tasks in a zero-shot manner or with lightweight finetuning. While the pretrained models are capable of many vision and vision-language tasks, we note that our models use the same pretraining data as previous methods [13, 21, 32, 33] and additional analysis of the data and the resulting model is necessary before the use of the models in practice. We show CoCa models are more robust on corrupted images, but it could still be vulnerable to other image corruptions that are not yet captured by current evaluation sets or in real-world scenarios. For both

loss	LE	FT
$\mathcal{L}_{\text{Cls}}$	81.0	85.1
$\mathcal{L}_{\text{Cap}}$	82.1	84.9

(a) 编码器-解码器模型与单编码器模型（在 JFT 上训练）。

loss	ZS	VQA	TPU cost
$\mathcal{L}_{\text{Con}}$	70.7	59.2	1×
$\mathcal{L}_{\text{Cap}}$	-	68.9	1.17×
$\mathcal{L}_{\textbf{CoCa}}$	**71.6**	**69.0**	**1.18**×

(b) 训练目标消融。

$\lambda_{\text{Cap}} : \lambda_{\text{Con}}$	ZS	VQA
1:1	71.5	68.6
1:2	71.0	68.1
2:1	**71.6**	**69.0**

(c) 训练目标权重。

n_{uni}	ZS	VQA
3	70.2	69.0
6	**71.6**	**69.0**
9	71.4	68.8

(d) 单模态解码器层。

variant	AE	MSCOCO
1 [CLS]	**80.7**	**41.4**
+ text tokens	80.3	40.2
8 [CLS]	80.3	36.9
+ text tokens	80.4	40.3

(e) 对比式文本嵌入设计消融。

variant	ZS	VQA
parallel	71.2	68.7
cascade	**71.6**	**69.0**
$n_{\text{query}} = 0$	71.5	69.0
$n_{\text{query}} = 1$	69.3	64.4
$n_{\text{query}} = 32$	71.2	68.2

(f) 注意力池化器设计消融。

表 8：CoCa 消融实验。在 ImageNet 分类上，我们报告以下设置的 top-1 准确率：零样本（ZS）、线性评估（LE）、在冻结特征上使用池化器的注意力评估（AE），以及微调（FT）。在 MSCOCO 检索上，我们报告图像到文本和文本到图像 R@1 的平均值。在 VQA 上，我们报告开发集 vqa 分数。默认 CoCa 设置以粗体显示。

我们发现，将解码器解耦为一半可以保持良好的平衡。一种可能的解释是，用于检索的全局文本表示不需要深层模块 [33]，而浅层的早期融合对于多模态理解也可能并非必要。接下来，我们探索提取单模态文本嵌入的各种选项。具体而言，我们实验了可学习 [CLS] 标记的数量以及聚合设计。对于后者，我们仅对 [CLS] 标记进行聚合，或对 [CLS] 与原始输入句子的拼接进行聚合。有趣的是，在表 8e 中，我们发现对于纯视觉和跨模态检索任务，训练单个 [CLS] 标记且不使用原始输入更为理想。这表明，学习一个额外的简单句子表示可以减轻对比损失与字幕生成损失之间的干扰，并且其能力足以实现强泛化。

注意力池化器。CoCa 在其设计中利用注意力池化器，既用于不同的预训练目标，也用于特定目标的下游任务适配。在预训练中，我们比较了几种使用池化器处理对比损失和生成损失的设计变体：（1）“并行”设计，即如图 2 所示，在 Vision Transformer 编码器输出上同时提取对比损失和生成损失；（2）“级联”设计，即将对比池化器应用于生成池化器的输出之上。表 8f 展示了这些变体的结果。经验上，我们发现，在小规模下，“级联”版本（将对比池化器置于生成池化器之上）表现更好，并在所有 CoCa 模型中默认使用。我们还研究了查询数量的影响，其中 $n_{\text{query}} = 0$ 表示不使用生成池化器（因此所有 ViT 输出标记都用于解码器交叉注意力）。结果表明，这两项任务都更偏好较长的、包含详细图像标记的序列，其代价是计算量和参数量略有增加。因此，我们使用长度为 256 的生成池化器，以提升多模态理解基准性能，同时仍保持强大的冻结特征能力。

5 更广泛的影响

本文提出了一种在网络规模数据集上进行图像-文本预训练的方法，能够以零样本方式或通过轻量级微调迁移到广泛的下游任务。尽管预训练模型能够完成许多视觉和视觉-语言任务，但我们注意到，我们的模型使用了与先前方法 [13, 21, 32, 33] 相同的预训练数据，因此在实际使用这些模型之前，有必要对数据及所得模型进行额外分析。我们表明，CoCa 模型在受损图像上更加鲁棒，但它仍可能容易受到当前评估集或真实场景中尚未涵盖的其他图像损坏的影响。对于两者

the data and model, further community exploration is required to understand the broader impacts including but not limited to fairness, social bias and potential misuse.

6 Conclusion

In this work we present Contrastive Captioners (CoCa), a new image-text foundation model family that subsumes existing vision pretraining paradigms with natural language supervision. Pretrained on image-text pairs from various data sources in a single stage, CoCa efficiently combines contrastive and captioning objectives in an encoder-decoder model. CoCa obtains a series of state-of-the-art performance with a single checkpoint on a wide spectrum of vision and vision-language problems. Our work bridges the gap among various pretraining approaches and we hope it motivates new directions for image-text foundation models.

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对于数据和模型，还需要社区进一步探索，以了解其更广泛的影响，包括但不限于公平性、社会偏见和潜在滥用。

6 结论

在这项工作中，我们提出了对比式字幕生成器（Contrastive Captioners，CoCa），这是一个新的图像-文本基础模型家族，它以自然语言监督涵盖了现有的视觉预训练范式。CoCa 在单阶段中基于来自各种数据源的图像-文本对进行预训练，并在编码器-解码器模型中高效结合了对比学习和字幕生成目标。CoCa 仅凭一个检查点，就在广泛的视觉及视觉-语言问题上取得了一系列最先进的性能。我们的工作弥合了各种预训练方法之间的差距，并希望能够激发图像-文本基础模型的新方向。

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A Visual Recognition Finetuning Details

Hyper-parameter	ImageNet		Kinetics-400/600/700		Moments-in-Time	
	Frozen-feature	Finetuning	Frozen-feature	Finetuning	Frozen-feature	Finetuning
Optimizer	Adafacter with Decoupled Weight Decay					
Gradient clip	1.0					
EMA decay rate	0.9999					
LR decay schedule	Cosine Schedule Decaying to Zero					
Loss	Softmax					
MixUp	None					
CutMix	None					
AutoAugment	None					
RepeatedAugment	None					
RandAugment	2, 20	2, 20	None	None	None	None
Label smoothing	0.2	0.5	0.1	0.1	0.0	0.0
Train steps	200k	200k	120k	120k	120k	120k
Train batch size	512	512	128	128	128	128
Pooler LR	5e-4	5e-4	5e-4	5e-4	5e-4	5e-4
Encoder LR	0.0	5e-4	0.0	5e-4	0.0	5e-4
Warm-up steps	0	0	1000	1000	1000	1000
Weight decay rate	0.01	0.01	0.0	0.0	0.0	0.0

Table 9: Hyper-parameters used in the visual recognition experiments.

In addition to zero-shot transfer, we evaluate frozen-feature and finetuning performance of CoCa on visual recognition tasks. For frozen-feature evaluation, we add an attentional pooling layer (pooler) on top of the output sequence of visual features and an additional softmax cross entropy loss layer to learn classification of images and videos. For finetuning, we adapt the same architecture as frozen-feature evaluation (thus also with poolers) and finetune both encoder and pooler. All learning hyperparameters are listed in Table 9.

B Multimodal Understanding Finetuning Details

Hyper-parameter	VQA	SNLI-VE	NLVR2	MSCOCO	NoCaps
Optimizer	Adafacter with Decoupled Weight Decay				
Gradient clip	1.0				
LR decay schedule	Cosine Schedule Decaying to Zero				
RandAugment	1, 10	1, 10	None	None	None
Train steps	100k	50k	50k	50k	10k
Train batch size	64	128	64	128	128
Pooler LR	5e-4	1e-3	5e-4	NA	NA
Encoder LR	2e-5	5e-5	2e-5	1e-5	1e-5
Warm-up steps	1000	1000	1000	1000	1000
Weight decay rate	0.1	0.1	0.1	0.1	0.1

Table 10: Hyper-parameters used in the multimodal experiments.

CoCa is an encoder-decoder model and the final decoder outputs can be used for multimodal understanding/generation. Thus, we evaluate on popular vision-language benchmarks. We mainly follow the same setup introduced in [16]. All hyper-parameters are listed in Table 10.

For multimodal classification, we feed the image into the encoder and the corresponding text to the decoder. We then apply another attentional pooler with a single query to extract embedding from the decoder output, and train a linear classifier on top of the pooled embedding. For VQA v2 [75], we follow prior work and formulate the task as a classification problem over 3,129 most frequent answers in the training set. We additionally enable cotraining with the generative loss on the concatenated pairs of textual questions and answers to improve model robustness. Similarly for SNLI-VE, the image and the textual hypothesis are fed to encoder and decoder separately, and the classifier is trained to predict the relation between them as entailment, neutral or contradiction. For NLVR2, we

视觉识别微调细节

Hyper-parameter	ImageNet		Kinetics-400/600/700		Moments-in-Time	
	Frozen-feature	Finetuning	Frozen-feature	Finetuning	Frozen-feature	Finetuning
Optimizer	Adafacter with Decoupled Weight Decay					
Gradient clip	1.0					
EMA decay rate	0.9999					
LR decay schedule	Cosine Schedule Decaying to Zero					
Loss	Softmax					
MixUp	None					
CutMix	None					
AutoAugment	None					
RepeatedAugment	None					
RandAugment	2, 20	2, 20	None	None	None	None
Label smoothing	0.2	0.5	0.1	0.1	0.0	0.0
Train steps	200k	200k	120k	120k	120k	120k
Train batch size	512	512	128	128	128	128
Pooler LR	5e-4	5e-4	5e-4	5e-4	5e-4	5e-4
Encoder LR	0.0	5e-4	0.0	5e-4	0.0	5e-4
Warm-up steps	0	0	1000	1000	1000	1000
Weight decay rate	0.01	0.01	0.0	0.0	0.0	0.0

表9：视觉识别实验中使用的超参数。

除了零样本迁移外，我们还评估了 CoCa 在视觉识别任务上的冻结特征和微调性能。对于冻结特征评估，我们在视觉特征输出序列之上添加一个注意力池化层（pooler），并额外添加一个 softmax 交叉熵损失层，用于学习图像和视频分类。对于微调，我们采用与冻结特征评估相同的架构（因此也包含 pooler），并同时微调编码器和 pooler。所有学习超参数列于表 9。

B 多模态理解微调细节

Hyper-parameter	VQA	SNLI-VE	NLVR2	MSCOCO	NoCaps
Optimizer	Adafacter with Decoupled Weight Decay				
Gradient clip	1.0				
LR decay schedule	Cosine Schedule Decaying to Zero				
RandAugment	1, 10	1, 10	None	None	None
Train steps	100k	50k	50k	50k	10k
Train batch size	64	128	64	128	128
Pooler LR	5e-4	1e-3	5e-4	NA	NA
Encoder LR	2e-5	5e-5	2e-5	1e-5	1e-5
Warm-up steps	1000	1000	1000	1000	1000
Weight decay rate	0.1	0.1	0.1	0.1	0.1

表 10：多模态实验中使用的超参数。

CoCa 是一种编码器-解码器模型，其最终解码器输出可用于多模态理解/生成。因此，我们在流行的视觉-语言基准上进行评估。我们主要遵循 [16] 中介绍的相同设置。所有超参数列于表 10。

对于多模态分类，我们将图像输入编码器，并将相应文本输入解码器。然后，我们应用另一个带有单个查询的注意力池化器，从解码器输出中提取嵌入，并在池化后的嵌入之上训练一个线性分类器。对于 VQA v2 [75]，我们遵循先前工作，将该任务表述为在训练集中 3,129 个最常见答案上的分类问题。我们还在文本问题与答案的拼接对上启用生成损失的联合训练，以提高模型鲁棒性。类似地，对于 SNLI-VE，图像和文本假设分别输入编码器和解码器，并训练分类器来预测它们之间的关系为蕴含、中立或矛盾。对于 NLVR2，我们

create two input pairs of each image and the text description, and concatenate them as input to the classifier. We do not use image augmentation for NLVR2.

For image captioning, we apply simple cross-entropy loss (same as the captioning loss used in pretraining) and finetune the model on the training split of MSCOCO to predict for MSCOCO test split and NoCaps online evaluation.

为每张图像和文本描述创建两个输入对，并将它们拼接后作为分类器的输入。我们不对 NLVR2 使用图像增强。

对于图像字幕生成，我们采用简单的交叉熵损失（与预训练中使用的字幕生成损失相同），并在 MSCOCO 的训练集上微调模型，以对 MSCOCO 测试集和 NoCaps 在线评估进行预测。